Progress Report

Project Title: ASU's Polymer-enhanced Cyanobacterial Bioproductivity (AUDACity)

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Section I. Executive Summary:

1. Technical Achievement(s) in Past Quarter

Major achievements of the project during the past quarter as reported herein include:

A) A series of polystyrene-based polymers was synthesized to try and reduce water uptake by varying the hydrophobicity of the linker between the hydrophilic quaternary ammonium ions that are critical for the moisture swing. The ion exchange capacity (IEC) of these materials (4.0 to 4.7 mmol/g) was significantly higher than that of a commercial A501 material (2.0 mmol/g) currently used for cultivating, which has a significant impact on the overall cost to deliver CO_2. Several polystyrene-based materials, including a more hydrophobic variant MVAM_105-2, were biocompatible and led to good growth—up to OD 5 over eight days—when cultivated with under alkaline conditions, either by adding 10 mM NaOH or 50 mM CAPS buffer.

B) The drying rate of a commercial resin was analyzed by thermogravimetric analysis and found to dry faster (39 vs 60 min) with an inactive chloride counter ion than the active bicarbonate ion, which we hypothesize is due to a higher hydration number and dissociation factor of the bicarbonate ion leading to more bonds with water.

C) Elemental analysis of the commercial Excellion and A501 resins revealed that anions from BG11 medium bound to these resins reduced the CO_2 binding capacity to 12.7% and 7.5% of the initial capacity, respectively, after eight soaks in fresh BG11 medium. Addition of 20 mM (moisture swing active) bicarbonate ions ($NaHCO_3$) to the BG11 medium enabled 66 and 60% of the CO_2 binding capacity to be retained for Excellion and A501 resins, respectively.

D) The AUDACity 1g laminar flow hood system with a belt made of Excellion strips was used to cultivate *Synechocystis* using four-hour cycles to reach OD 0.75 after eight days. The medium was found to retain ~10 mM bicarbonate that likely helped the Excellion belt retain 80% of its initial capacity after 10 days. The 1g system has been upgraded to include a more stable chain and tensioning system and drive motor for more reliable operation, a four-axle geometry to enable longer sorbent drying times, and a new belt with our most promising commercial sorbent A501 loaded in elongated mesh tubes, called pixie sticks, which will be evaluated next quarter.

E) The wind tunnel was used to measure sorbent drying and CO_2 absorption after the sorbent was desorbed in aqueous solutions of increasing alkalinity; solutions containing only deionized (DI) water or 10 mM Na_2CO_3 only partially unloaded the sorbent whereas a solution with 100 mM Na_2CO_3 completely unloaded the sorbent leading to CO_2 adsorption of 670 µmoles CO_2/g sorbent, which is the sorbent's full theoretical capacity. The wind tunnel was also used to analyze sorbent drying and CO_2 absorption as a function of wind velocity, where velocities of 2.5 and 5 m/s dried and fully loaded the sorbent within 1.5–2.0 hours, whereas at a velocity of 0.7 m/s the drying and full loading took 3.5–4.0 hours. The pixie stick form factor was found to dry and load the A501 sorbent 6-fold faster than when the sorbent was contained within two more densely packed mesh bags.

F) Techno economic analysis found that the average sorbent drying/loading rate and a sorbent's IEC have a large impact on the cost of CO_2 delivery and are keys to achieving future scenarios of reducing the cost of CO_2 delivered from ~$300/tonne to $50/tonne.

G) The AUDACity 100g system design was updated to include a double-loop form factor that will reduce power usage and shading of the pond and to accommodate a double-wide sorbent packet.

H) *Synechocystis* strains that had been selected for growth in the presence of methacrylate polymers for over 100 generations (38 subcultures) were found to maintain their ability to grow well in the presence of methacrylate polymers after 10 generations without any selective pressure, indicating genetic changes made during selection are stable. These strains will be sequenced during the next quarter to identify these genetic changes and the strains will be used for cultivation with other polymers containing quaternary ammoniums ions.

I) A fluorescence-activated cell sorting system was used to isolate particles of phycocyanin encapsulated with milk fat globule membranes (MFGM) with size ranges of 1 – 10 µm, which maintained 75% of their color after being subjected to heat treatment (100 °C for 10 minutes).

2. Project Schedule Status

Task 1 (Polymer design and synthesis) is behind schedule. However, several polystyrene-based materials have been identified that are biocompatible and lead to good growth, including some that are more hydrophobic than our best commercial A501 sorbent, which should reduce water losses and drying/loading times. These materials also have higher IECs than A501, which should improve the amount of CO_2 captured per day and overall economics.

Task 2 (Polymer biocompatibility and lab-scale DAC cultivation of *Synechocystis*) is largely on schedule. Work continues to evaluate additional synthesized polymers and identify optimal cultivation conditions with commercial and synthesized polymers.

Task 3 (Design, develop and evaluate AUDACity systems) is behind schedule. The AUDACity 1g laminar flow hood system was recently upgraded and now contains a belt made of a commercial A501 sorbent with elongated mesh tubes we plan to scale up for the AUDACity 100g system. The AUDACity 1g rooftop system is operational being used to test concepts for the 1g and 100g systems. The 100g system design has been updated and costed out and part orders have been submitted to build to first 100g system.

Task 4 (AUDACity outdoor cultivation trials at AzCATI) is a little behind schedule; the TERA application has been completed and will be submitted early next quarter. We expect the TERA to be approved within 30 days in line with previous similar applications.

Task 5 (Techno-Economic and Life Cycle Analyses) is on schedule.

3. Project Budget Status

Table A shows the spending for the quarter and fiscal year to date compared to what was planned in the budget and the corresponding variance. Current spending in FY23 Q3 is below budget, but we expect a significant increase next quarter as we build the AUDACity 100g system. Cost share contributions now make up 22% of total expenditures. **Table B** shows the DOE funds received in the quarter and fiscal year to date.

Table A. Variance Reporting

Period	Spending Plan	Actual Costs	Variance
FY23, Q3	$234,964	$179,482	-24%
FY23, YTD	$704,895	$617,035	-12%

Table B. Funding Received

Period	Funds
FY23, Q3	$0
FY23, YTD	$179,909

4. Relevance to Programmatic Goals

The Algae Bioproducts and CO_2 Direct-Air-Capture Efficiency (ABCDE) program is focused on decoupling microalgae cultivation from concentrated CO_2 sources, increasing biomass productivity and revenue potential while maintaining biomass quality and reducing the cost of CO_2 delivered to algal systems. AUDACity uses low-cost DAC polymers to capture and deliver CO_2 from air to cultivate genetically modified cyanobacteria that excrete laurate or methyl laurate as a fuel and chemical feedstock and also contain the high-value product phycocyanin (PC). This will enable AUDACity to help DOE achieve the stated goals of the ABCDE program.

Project results to date demonstrate progress toward meeting these programmatic goals including **1)** cultivating cyanobacteria using 100% CO_2 delivered by DAC polymers with an average productivity of ~186 mg L^{-1} d^{-1} over six days and identifying laboratory conditions to enable long-term growth of a fast-growing cyanobacterium to increase areal productivity from DAC, **2)** developing low-cost DAC polymers with rapid loading and unloading rates projected to cost ≤ $100/tonne CO_2 delivered from air to the cyanobacterial system, **3)** identifying a path to producing $2.50/GGE of biofuel using high-value products (phycocyanin, protein) to offset the cost of cultivation, **4)** identifying a pathway to reduce the minimum fuel selling price of extracellular harvested methyl laurate of $5.80/GGE from $7.45/GGE for conventional biomass processing to fuel, **5)** establishing a baseline biomass composition with 10 wt% (methyl) laurate and 12–17 wt% PC and low-cost procedures to isolate PC with a color number of 500 (analytical grade corresponds to PC with a color number >400) to ensure a high value of the biomass, and **6)** identifying conditions to stabilize PC at 100 ºC for longer periods of time in dry form and at 100 ºC in water when encapsulated in liposomes or lipid vesicles to increase the biomass revenue potential.

5. Comparison to the State-of-the-Art

Several DAC technologies are being developed to capture and concentrate CO_2 from air, but economic cost projections of these technologies are in the range of $95–230/tonne CO_2. AUDACity can greatly reduce CO_2 capture costs by eliminating forced air movement, replacing costly cooling towers with low-cost polymer-based materials, and eliminating caustic liquid sorbents (e.g., KOH) and high-temperature calcination. Water evaporation replaces heat as the source of exergy to concentrate the CO_2. Further, by

using the culture medium to directly desorb and deliver CO_2, AUDACity avoids the costly and kinetically slow steps of extracting, compressing, delivering and dissolving CO_2 gas.

Results to date from this project show that synthesized DAC polymers are able to deliver ~90% of their CO_2 after being submerged in water for 15 minutes and are able to load with CO_2 to 90% of their capacity within 1.5 hours in a wind tunnel blowing at 5 m/s. These resin materials capture about 3% of their weight in CO_2 and are very inexpensive, which should enable capture and concentration of atmospheric CO_2 for < $100/tonne CO_2.

(Methyl) laurate-producing cyanobacteria have lower biomass productivity in ponds than typical green algae strains (e.g., *Scenedesmus*), but the methyl laurate produced significantly improves the fuel yield of the biorefinery to 259 GGE/tonne over 113 GGE/tonne for *Scenedesmus*. Further, significant value produced from cultivating the high-value co-product phycocyanin can offset the entire cultivation costs while providing "free biomass" for HTL conversion into fuel at $2.50/GGE to meet DOE targets. We are further working on improving the thermostability of phycocyanin that will enable new PC markets that would allow for cost-effective biofuel production at even larger scales (beyond saturating the current PC market).

6. Lessons Learned

1) Adding 20 mM bicarbonate to BG11 helps retain 60 to 66% of the CO_2 binding capacity of A501 and Excellion ion exchange resins, respectively, after eight wet/dry cycles.

2) Loading powdered A501 sorbent within elongated mesh tubes helps the sorbent dry and load CO_2 six-fold faster than when tightly packed in mesh bags.

3) Hydrophobic coatings applied to the mesh reduced the water uptake after soaking by 150% and decreased the drying time from about 2 hours to 45 minutes.

4) Solutions containing only deionized (DI) water or 10 mM Na_2CO_3 only partially unloaded the sorbent whereas a solution with 100 mM Na_2CO_3 completely unloaded the sorbent leading to CO_2 absorption of 670 µmoles CO_2/g sorbent as measured in the wind tunnel, which is the sorbent's full theoretical capacity.

5) Techno-economic analysis found the average sorbent drying/loading rate and a sorbent's IEC have a large impact on the cost of CO_2 delivery and are keys to achieving future scenarios that reduce the cost of CO_2 delivered from ~$300/tonne to $50/tonne.

Section II. Task Updates

Task 1: Polymer design and synthesis (Months 4–36)

1. Planned Activities:

1. Synthesize biocompatible styrene-based polymers
2. Investigate alternative routes to produce membranes with higher capturing capacity and pH stability
3. Possibly tune the hydrophobic properties of the most promising biocompatible polymers
4. Perform Thermogravimetric Analysis-Sorption Analysis (TGA-SA) on the most promising polymers to measure their water uptakes and drying times
5. Analyze coating retention after water soaks using IR spectroscopy
6. **SMART Milestone 1.2.1** will be reported for selected relevant polymer sorbents

SMART Milestone 1.2.1: Processed Gen 1 sorbent captures ≥ 2.0 μmol CO_2 s^{-1} g^{-1} sorbent with ≤ 1 h cycle time on lab scale (M15)

Deliverable 1.3.3: Determine conditions to produce a sorbent-coated belt with Generation 2 sorbent (M18).

2. Accomplishments:

Polystyrene-based polymer synthesis

Polystyrene-based polymers synthesized in the previous quarter, which mimic some aspects of commercial anion exchange resins such as A501, were found to be significantly more stable under alkaline conditions than previous methacrylate-based polymers and to be biocompatible, as described in the FY23-Q2 report (under Task 1). However, the water uptake of these materials was still quite substantial, much like the commercial anion exchange resins, leading to long drying and loading times and excess water loss. Therefore, a series of polystyrene-based polymers was synthesized to increase hydrophobicity and reduce water uptake by varying the hydrophobicity of the linker between the hydrophilic quaternary ammonium ions that are critical for the moisture swing. The linker consisted of either aliphatic hydrocarbon chains of 2 (CL_3) or 6 (CL_1) carbons in length or a phenyl group (CL_2), as shown in **Figure 1.1**. These polymers were synthesized by mixing a solution of monomer(s) (VBTEA, DVB, CL_1, CL_2 or CL_3) (30 or 50% m/m) in solvent with 4% mol of photoinitiatior and poured into petri dishes. The polymerization was initiated by exposure to UV light (365 nm) for 30 minutes. The thickness of the polymers was fixed at 1 ± 0.5 mm. The polymer was washed several times using a MeOH:water mixture to remove excess solvent and unreacted monomer, and then rinsed for 2 h in a mesh bag in deionized (DI) water. The final polymer was dried overnight under vacuum at 50 °C. The moisture-swing-inactive chloride counterion of the quaternary ammonium used during synthesis was exchanged with the active bicarbonate counter ion by placing the polymer in a mesh bag and soaking in 1 M $KHCO_3$ in water for 24 h. This process was repeated twice to reach complete exchange of chloride with the bicarbonate counterion. After complete conversion, the polymer was rinsed in DI water to remove any remaining salt and dried in

ambient air. The ion exchange capacity of the polymers was determined experimentally by the concentration of chloride in solution using Mohr's Method, by precipitation of silver chloride.

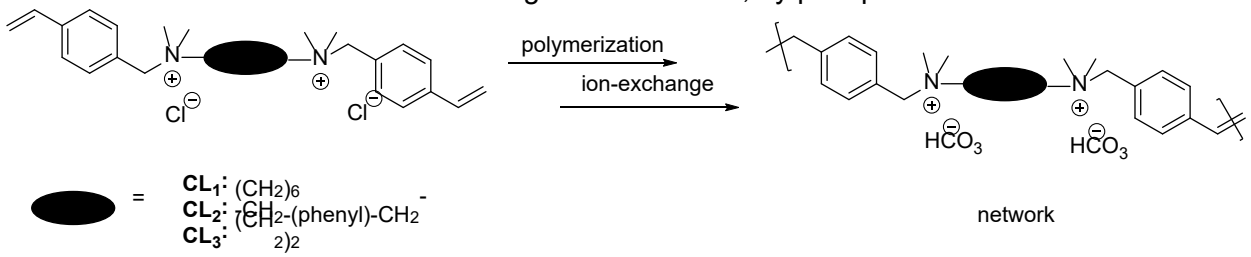


Figure 1.1. Synthesis of polystyrene-based polymers with quaternary ammonium groups in the chains.

Quantifying ion-exchange capacity by titration.

The ion-exchange capacity sets the upper limit of the CO_2-binding capacity for a given polymer. A high ion-exchange capacity is critical for achieving cost-effective CO_2 capture and delivery to the cyanobacteria. **Table 1.1** summarizes the titration data collected for a variety of synthesized styrene-based polymers used to quantify their ion-exchange capacity, as described in the FY23-Q2 report (under Task 1). In addition to titration, we quantified the amount of exchanged chloride using QuanTab chloride strips. The data demonstrate a close alignment between the capacities measured via the titration method and the QuanTab strips, thereby underscoring the accuracy of Mohr's titration method. The theoretical IEC for the commercial ion-exchange polymer A501, as listed on the Purolite website, is 0.5 meq/L or 1.3 mmol/g as per **Equation 1.1**. Moreover, we normalized our titrations using the dry mass of the resin. Given the high water content of the A501 resin, we initially stabilized its weight by drying the sample in ambient air for 24 hours. Our observations show that the resin undergoes a mass loss of approximately 60% after drying for this period. The dry mass of these samples was then used to calculate the experimental IEC. Furthermore, our data indicate that the commercial resin displays a significantly lower capacity compared to all the synthesized polymers.

$$Theoretical\ IEC = \frac{0.5\ meq}{L} x \frac{1\ mmol}{1\ meq} x \frac{L}{385\ g} = 1.3 \frac{mmol}{g} \quad \text{(Equation 1.1)}$$

Table 1.1: Ion exchange capacity (IEC) of commercial and synthesized polymers measured by titration and with QuanTab strips.

Sample Name	Polymer Mass [g]	Volume of 1M $AgNO_3$ added [ml]	Measured IEC - titration [mmol/gr]	Measured IEC – QuanTab Strips [mmol/gr]	Theoretical IEC [mmol/gr]
A501P (commercial)					
A501P	0.849	2	2	2	1.3
MAVM82 (CL1 with 6C linker)					
MVM94-1 [2 wt% DMPA, 30% monomer concentration]	2.6747	11	4	3.3	4.2
MVM94-2 [2 wt% DMPA, 50% monomer concentration]	2.3410	4.9	2.1	2.3	4.2
MVM94-3 [1 wt% DMPA, 30% monomer concentration]	4.3050	8.3	2	2.6	4.2
MAVM105 (CL2 with Phenyl linker)					
MVM105-1 [30% monomer concertation]	1.8	3.6	2	2.2	4
MVM105-2 [50% monomer concertation]	0.8	3.0	3.8	3.6	4
MAVM109 (CL3 with 2C linker)					
MVM109	3.37	10	3	2.9	4.7
MAVM 6A(CL1+DVB+VDTEA)					
MVM6A	2	6.4	3.2	-	4.2
MAVM 5B(CL1+DVB)					
MVM 5B	0.76	3.4	4	-	4

Quantifying sorbent drying:

While the anion exchange polymers utilized in AUDACity generally have very fast CO_2 delivery, they tend to absorb significant quantities of water and thus take a long time to dry and load with CO_2. The drying rate is influenced by many factors such as temperature, wind speed and relative humidity, which are difficult to control in the open laboratory environment, making comparison between various sorbent materials challenging. The drying time of the commercially exchanged resin A501 was measured using a TA Thermogravimetric Analyzer (TGA 5500), which is able to control temperature and maintains a constant wind speed. A501 resin with the inactive chloride counterion was used as the baseline, and the resin with bicarbonate was prepared by immersing the resin in a 0.1 M potassium bicarbonate solution for 30 minutes. Subsequently, the samples were removed from the solution via gravitational filtering to remove excess water and loaded into the TGA. During the initial heating phase, the temperature of the samples was increased to 50 °C at an increment of 10 °C per minute, all within a nitrogen atmosphere. The temperature was maintained at 50 °C for 180 minutes. **Figure 1.2** shows the sample weight change as a function of time as collected by the TGA and **Table 1.2** summarizes the drying time for both chloride and bicarbonate counter ions as well as the drying rates as calculated using **Equation 1.2**.

$$Drying\ rate = \left(\frac{change\ in\ mass\ (mg)}{change\ in\ time\ (min)}\right) \qquad \text{(Equation 1.2)}$$

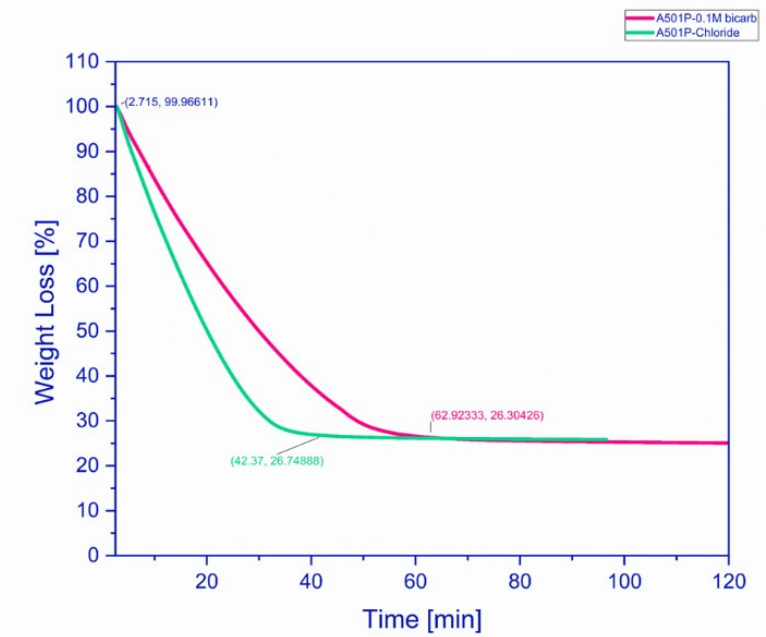


Figure 1.2: Weight loss as a function of time for A501P resin with chloride counter ion (green) and bicarbonate counter ion (pink) as collected by TGA.

Table 1.2: Comparison of drying times for commercial resin in chloride and bicarbonate forms

Sample Name	Weight loss [%]	Drying time [min]	Drying Rate [mg/min]
A501P-Chloride	73.3	39	0.21
A501P-Bicarbonate	73.7	60	0.17

The collected results showed that the drying time of the resin with bicarbonate counterion was about 70% higher than that with the chloride counterion. This could be due to the bicarbonate ions having the tendency to absorb more water molecules than chloride. This is due to a several factors such as hydration number and dissociation factor. The hydration number is defined by the number of water molecules that an ion can attract. Chloride ions can bond with about 4 to 9 water molecules whereas bicarbonate can bond with 7 to 8 water molecules; thus bicarbonate has the potential of forming 3-4 more hydrogen bonds with surrounding water molecules than the chloride ions. From the TGA data, we observed a shorter drying time for the chloride counterion indicating that in our experimental matrix the chloride is bonding with fewer water molecules than bicarbonate. Additionally, bicarbonate ions dissociate more than chloride resulting in higher water absorbance (Amel, 2016). The dissociation factor is important because it affects the number of free ions in the solution. These free ions can then attract and bond with more water molecules, which in turn increases the water uptake of the substance and influences drying times.

3. Explanation of Variance:

In the previous quarter, our focus was primarily on two objectives. The first involved the synthesis of biocompatible styrene-based quaternary ammonium polymers with enhanced hydrophobic properties. We explored alternative routes to improve hydrophobicity, such as varying the linker length between molecules, keeping biocompatibility of the polymers as this is a crucial property for our project. The balance between hydrophobicity and biocompatibility has proven to be a challenging aspect that we plan to address further in subsequent stages of the project. We have also considered alternative routes to improve the overall stability and performance of the polymers. Innovative grafting techniques were explored, involving gamma irradiation on partially fluorinated sheets, with the ultimate aim of grafting monomers capable of carbon capture while preserving essential biocompatibility. Despite some unknown challenges that may require further research, we remain optimistic about the potential of these materials as secondary resources. As some activities from our previous tasks are still pending, we are committed to accomplishing these in the next quarter. A plan for the next quarter, which includes these activities, is outlined in the following section.

4. Plans for Next Quarter:

Next quarter we will:

1. Synthesize biocompatible styrene-based polymers while increasing the hydrophobic properties
2. Investigate alternative resins by grafting quaternary ammonium functional groups onto partially fluorinated sheets using gamma radiation
3. Perform Thermogravimetric Analysis-Sorption Analysis (TGA-SA) on the most promising polymers to measure their water uptake and drying times
4. Analyze coating retention after water soaks using IR spectroscopy.
5. **SMART Milestone 1.2.1** will be reported for selected relevant polymer sorbents

SMART Milestone 1.2.1: Processed Gen 1 sorbent captures ≥ 2.0 μmol CO_2 s^{-1} g^{-1} sorbent with ≤ 1 h cycle time on lab scale (M15)

Deliverable 1.3.3: Determine conditions to produce a sorbent-coated belt with Generation 2 sorbent (M18).

Task 2: Polymer biocompatibility and lab-scale DAC cultivation of *Synechocystis* (M4–36)

1. Planned Activities:

<u>Biocompatibility</u>

- Test biocompatibility of polystyrene-based polymers with *Synechocystis* cultures and monitor the stability and performance of the polymer upon repeated use in *Synechocystis* cultures.
- Sequence the adapting *Synechocystis* strains after the strains have evolved permanent tolerance to the methacrylate polymer to determine common genetic changes in the three parallel cultures.
- Use the adapting strains with improved methacrylate tolerance to test biocompatibility of newly synthesized polymers, as needed.

AUDACity 1g cultivation

Work in the coming quarter will focus on characterizing growth with the (uncoated) Excellion belt and we may explore adding a Starbrite hydrophobic coating to reduce water uptake. We will also begin cultivation trials using with the most promising powdered commercial polymer (A501) contained within the mesh pixie stick design once it has been evaluated abiotically in the second 1g system. Furthermore, by extending the belt to increase drying time, we plan to be able to meet the growth targets laid out in Milestone 2.2.1 under continuous and diurnal light conditions.

Phycocyanin

We will quantify the encapsulation process to encapsulate phycocyanin in other lipid systems (milk fat globule membranes, archaeosomes) that are more thermostable.

Outstanding Deliverables/Milestones:

Deliverable 2.2.4: A) Determine culture parameters (including polymer-related ones) that maximize growth and bioproducts when C_i is provided by polymers, B) measure PC color number and yield as well as stability to temperature, pH and light, and C) carbon utilization efficiency (M21).

Deliverable 2.2.5: Define conditions that, in the laboratory under diurnal conditions, lead to the most economic production of biomass and bioproducts (M24).

Milestone 2.2.3: Lab-scale demonstration of AUDACity *1g* system for cyanobacterial cultivation meeting technical targets in Milestone 2.2.1 (M24).

2. Accomplishments:

Biocompatibility.

Deliverable 2.2.4: A) Determine culture parameters (including polymer-related ones) that maximize growth and bioproducts when C_i is provided by polymers.

A1. Elemental analysis of N, Cl and P remaining in BG-11 after soaking with an Excellion strip and Amberlite A501 resin

The major issue with sorbents we have encountered in cyanobacterial cultivation is anion depletion from growth medium by sorbents, which resulted in poor growth of *Synechocystis* cultures after a couple of soaks with sorbents. In addition, binding of anions to the sorbent leads to reduced CO_2 binding capacity of sorbents.

To determine whether the sub-optimal growth of cultures that had been exposed to sorbents was due to ion depletion from the medium (with ions adsorbing to the polymer), we carried out an elemental analysis (detecting N, Cl and P using a Lachat QC 8000 instrument) on BG11 medium after exposure to the polymer (either Excellion or A501) for 1-8 times, with or without addition of 20 mM $NaHCO_3$, which is a moisture-swing-active anion added to compete with moisture-swing-inactive anions from the BG11 medium binding to the polymer. As shown in **Tables 2.1 and 2.2**, we observed that indeed the BG11 ions were depleted from the medium after soaking with either Excellion or A501, and that after soaking the polymer multiple times in the medium anion-binding sites on the polymer were saturated and CO_2-binding and release efficiency had been reduced concomitantly. The elemental analysis showed that addition of 20 mM bicarbonate to BG11 during soaking did help prevent adsorption of NO_3^-, Cl^- and HPO_4^{2-} to the sorbent. After soaking in BG11 eight times in the presence of bicarbonate, the sorbents retained over 60% of their original CO_2 binding capacity, whereas sorbents retained only about 10% of their original CO_2-binding capacity when eight soaks in BG11 medium were carried out in the absence of bicarbonate.

Table 2.1. Elemental analysis of N, Cl and P remaining in BG-11 after soaking with an Excellion strip for the number of times indicated

	NO_3^- (mM)		Cl^- (mM)		PO_4^{3-} (µM)		Retained CO_2 release capacity %	
BG11 before soaking with strip	18.7		0.46		184			
After presoaking strip in BG11 with/without bicarb	BG11	BG11 + 20 mM bicarb	BG11	BG11 + 20 mM bicarb	BG11	BG11 + 20 mM bicarb	BG11	BG11+ 20 mM bicarb
1 time	3.6 (19%)	4.8 (26%)	0.16	0.23	40.5	58.4	88.2	
2 times	5.2 (28%)	9.4 (50%)	0.17	0.33	39.8	93.2	86.4	
4 times	13.0 (70%)	17.9 (96%)	0.49	0.41	85.3	138.6	38.6	65.5
8 times	19.0 (102%)	19.0 (102%)	0.42	0.43	145.6	165.7	12.7	66.0

Table 2.2. Elemental analysis of N, Cl and P remaining in BG-11 after soaking with 1 g A501 for the number of times indicated

	NO_3^- (mM)		Cl^- (mM)		PO_4^{3-} (µM)		Retained CO_2 release capacity %	
BG11 before soaking with A501	18.7		0.46		184			
After presoaking 1 g A501 in BG11 with/without bicarb	BG11	BG11 + 20 mM bicarb	BG11	BG11 + 20 mM bicarb	BG11	BG11 + 20 mM bicarb	BG11	BG11+ 20 mM bicarb
1 time	0.6 (3%)	2.7 (14%)	0.45	0.65	25.5	142.6	100	
2 times	1.3 (7%)	10.8 (58%)	0.32	0.62	48.1	147.7	98	
4 times	5.7 (30%)	16.3 (87%)	0.74	0.48	160.5	138.3	82	77
8 times	19.0 (101%)	18.9 (101%)	0.6	0.61	181.5	184	7.5	60

A2. Biocompatibility and effectiveness of polystyrene-based polymers in *Synechocystis* cultures.

A set of new polystyrene-based polymers, MVM_82 series 82_1, 82_2 and 82_3 was synthesized with different amounts of initiator and monomer (*N*,*N*,*N*,*N*-tetramethyl-*N*,*N*-bis(4-vinylbenzyl)hexane-1,6-diaminium dichloride). They had adequate CO_2 release capacity (14.7-15.2 mg CO_2/g), and were tested in *Synechocystis* cultures this quarter. Most synthesized polymers are expected to bind anions as the polymers contain quaternary ammonium ions. To reduce binding of anions (NO_3^-, Cl^- and HPO_4^{2-}) in BG11 medium to the sorbent, which would reduce the polymer's CO_2-binding effectiveness and remove critical minerals from the growth

medium, the three polystyrene polymers were pre-equilibrated in 0.5 L 20 mM bicarbonate in BG11 medium overnight in order to allow the polymer to equilibrate with the anion mixture in the BG-11 with bicarbonate media to prevent additional nutrient removal during cultivation. The polymers were then dried and used for cultivation of *Synechocystis*. Upon cultivation, the sorbent was soaked in BG11 in the presence of 20 mM bicarbonate for 30 min before inoculation and then *Synechocystis* cells were added after the sorbent was removed.
The results of cyanobacterial culturing are shown in **Figure 2.1**. All three sorbents performed similarly and supported *Synechocystis* to grow to an OD about 1.5-2 in 2 days. Then the growth slowed down, which could be due to the deficiency of anions in the culture that had been bound to the sorbent and thus removed from the medium. Addition of a new dose of BG11 nutrients on day 3 had little effect on the recovery of growth. With most polymers we observed that cultures grew to an OD_{730} of about 2-2.5, which is adequate but not stellar. Note that in contrast to methacrylate polymers that were largely not biocompatible at alkaline pH, all three polystyrene polymers appeared to be non-toxic as all three cultures grown with polystyrene polymers looked green and healthy during the entire growth period.

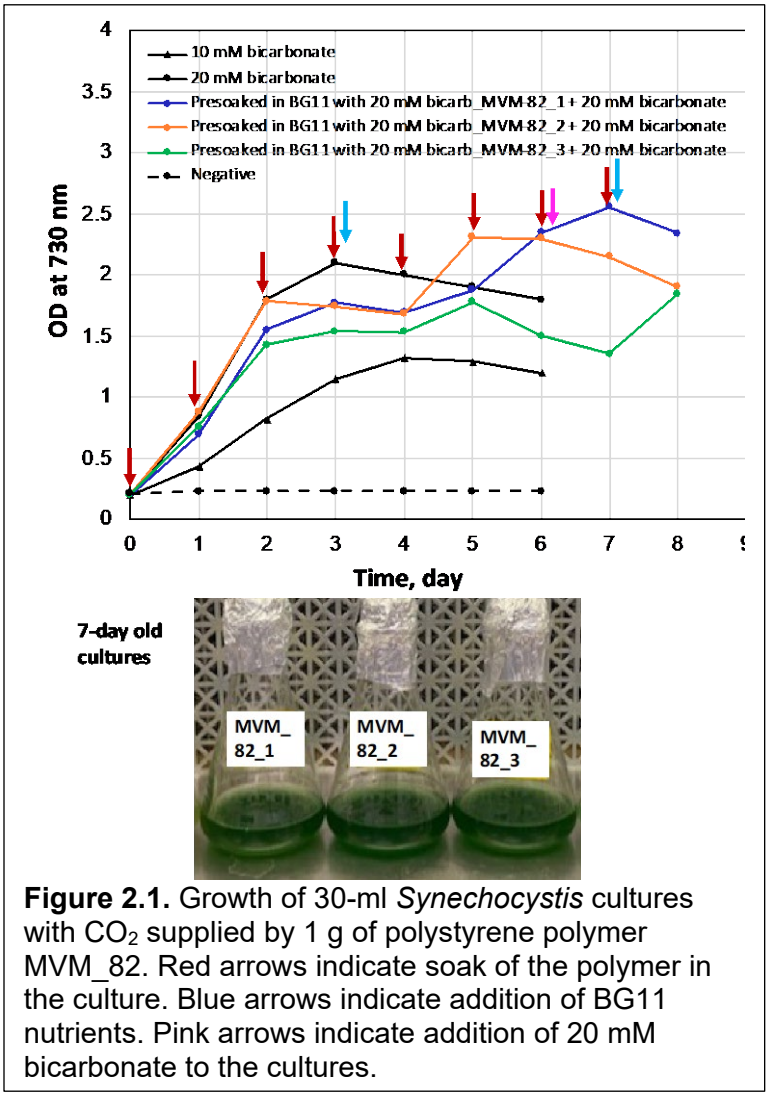


Figure 2.1. Growth of 30-ml *Synechocystis* cultures with CO_2 supplied by 1 g of polystyrene polymer MVM_82. Red arrows indicate soak of the polymer in the culture. Blue arrows indicate addition of BG11 nutrients. Pink arrows indicate addition of 20 mM bicarbonate to the cultures.

A potential reason for sub-optimal growth of the cultures, besides anion depletion from BG11, could be limited C_i availability in the medium due to relatively low pH: CO_2 is released from the sorbent, and this will decrease the pH. Therefore, we soaked the sorbent in BG11 at high pH (11.5). In addition to MVM_82, two other polystyrene polymers—a more hydrophobic MVM_105 and a more hydrophilic MVM109 (**Figure 2.2**)—were soaked for 30 min in sterile DI-H_2O that had been brought to pH 11.5 with 10 mM NaOH. Then the sorbents were removed from the alkaline water and BG11 nutrients and *Synechocystis* cells were added. As shown in Figure 2.2, the culture with polymer 94_2 (same as MVM_82 described earlier) grew up quickly in the first three days, and reached an OD of 2 by day 3. Growth was slowed down after that, consistent with other cultures grown with sorbents. After giving a second dose of BG11 nutrients on day 6 the culture started growing again and reached an OD of close to 4, suggesting that deficiency of BG11 nutrients removed

by sorbents, and not a lack of C_i, was limiting *Synechocystis* growth. The culture grown with 94_2 looked healthy during the entire growth period.

Polymers 109 and 105 were not biocompatible in this experiment. The culture with polymer 109 did not grow at all, and the culture with polymer 105 grew a little on day 1 and then stopped growing.

Identifying which anions are depleted by the sorbent and the frequency with which they would need to be added is important from a TEA perspective. The key nutrients in BG11 are nitrate and phosphate. A lack of fixed-nitrogen leads to a chlorotic phenotype, which was not observed; therefore, we hypothesized that phosphate was the main nutrient being depleted by soaking with polymer. Therefore, on a regular basis after starting the culture we provided cultures with extra phosphate rather than all of the BG11 nutrients. In addition, in order to maintain a consistently high alkaline pH during the entire growth period that facilitates CO_2 release from the sorbent, 50 mM CAPS was added to the cultures as CAPS has a pKa of 10.4 at 25 ºC that provides a buffering pH range of 9.7-11.1. For this experiment the sorbents of the three series (94, 105 and 109) were soaked in 50 ml DI-H_2O in the presence of 50 mM CAPS buffer (pH 10.5, NaOH) for 30 min and BG11 nutrients and cells were added after removal of sorbent. Sorbent 94_2 was also soaked in 50 ml DI-H_2O (pH 11.5, due to addition of 10 mM NaOH) for comparison.

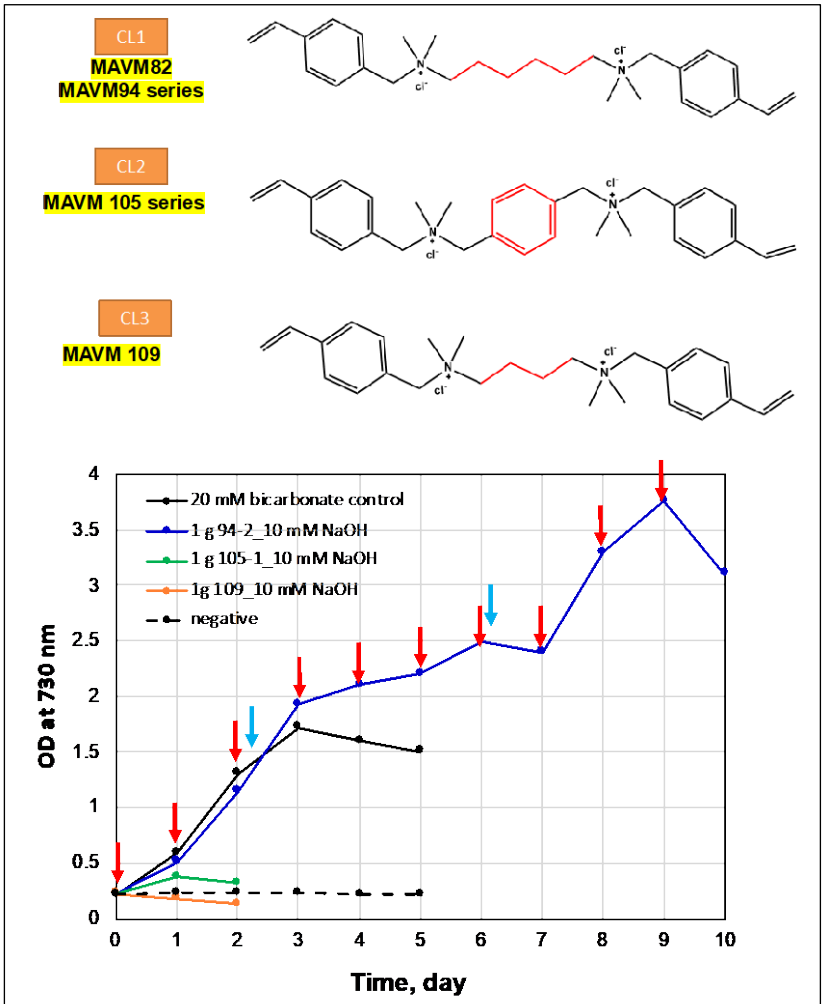


Figure 2.2. Structure of polystyrene polymers, MAVM 94 series, MAVM 105 and MAVM 109 series (top), and growth of 50-ml *Synechocystis* cultures with CO_2 supplied daily by 1 g of polystyrene polymer (bottom). Red arrows indicate times of soaking of the polymer in the culture. Blue arrows indicate addition of BG11 nutrients.

The results of this experiment are shown in **Figure 2.3**. Addition of phosphate on day 2 supported growth in all three cultures that were soaked with sorbent daily. However, addition of more phosphate to the cultures on day 4 did not help, suggesting that besides phosphate other components such as trace minerals or iron ammonium citrate may be depleted from the growth medium since the cultures did not show a chlorotic phenotype and appeared healthy. Indeed,

providing an additional dose of BG11 nutrients on day 5 led to additional growth in several cultures.

The two cultures grown with sorbent 94_2 grew well although the unbuffered one supported growth better. This fits with earlier experiments where we observed that strongly buffered media at high pH do not support *Synechocystis* growth well, presumably because of having to maintain an internal pH in their cells that is close to neutral. Also, the sorbent 105_1 that did not support growth in an earlier experiment now supported almost as much growth as sorbent 94_2. We hypothesize that toxic compounds were excreted from the polymer in first soaks and were no longer present in subsequent soaks. Nonetheless, cultures with sorbents 105_2 and 109 did not grow at all, suggesting that these two polymers still are not biocompatible upon second use.

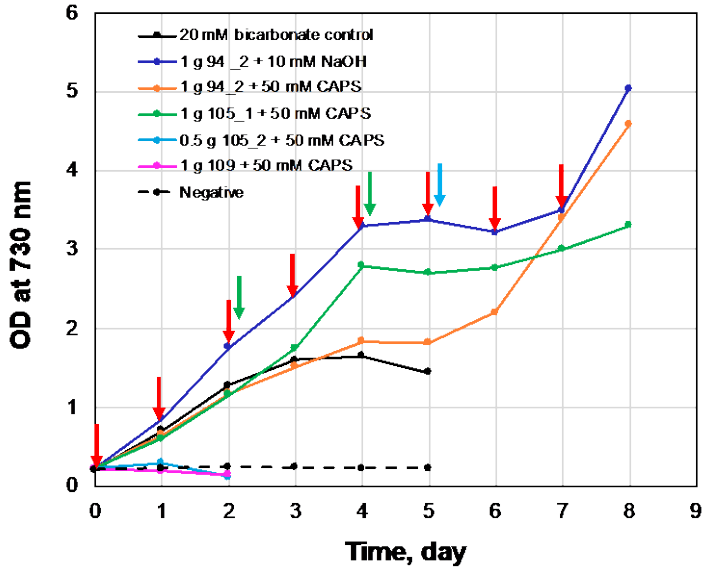


Figure 2.3. Growth of 50-ml *Synechocystis* cultures with CO_2 supplied by 1 g of polystyrene polymers that were soaked in DI-H_2O with 50 mM CAPS (pH 10.5, NaOH) for 30 min. Red arrows indicate soak of the polymer in the culture. Blue arrows indicate addition of BG11 nutrients. Green arrows indicate addition of phosphate to the culture.

In future experiments, we will try to identify more precisely which BG11 components specifically are depleted by incubation with the sorbent as this will have relevance for TEA analysis as well for understanding how to minimize decrease of sorbent functionality.

A3. *Synechocystis* strains evolving tolerance to the methacrylate polymer

Adaptive strain evolution has continued this quarter to generate cyanobacterial strains that are tolerant to the otherwise highly toxic methacrylate polymers. Three parallel *Synechocystis* cultures have been evolved for over 100 generations selecting for improved tolerance to the methacrylate polymer. Indeed, we developed strains with excellent tolerance in that a methacrylate polymer in powder form (<35 µm particles) that kills the unevolved control strain after soaking 1 g polymer in BG11 was tolerated well by all three strains. As shown in **Figure 2.4**, the three parallel cultures grown in BG11 soaked in 0.25-0.75 g of the usually highly toxic powder for 60 minutes had doubling times of less than a day without lag phase; moreover, the 24-h cultures looked green and healthy.

To confirm that the *Synechocystis* strains have developed lasting tolerance to the methacrylate polymer, one of the strains was grown without selective pressure (i.e., without methacrylate) for over 10 generations and then was tested for tolerance to the methacrylate polymer that had been soaked in BG11 for a day. As shown in **Figure 2.5**, the control strain turned bluish in 24 h and then bleached, whereas the evolved strain with or without having been grown for over 10

generations in the absence of methacrylate grew up nicely and was tolerant to the methacrylate polymer.

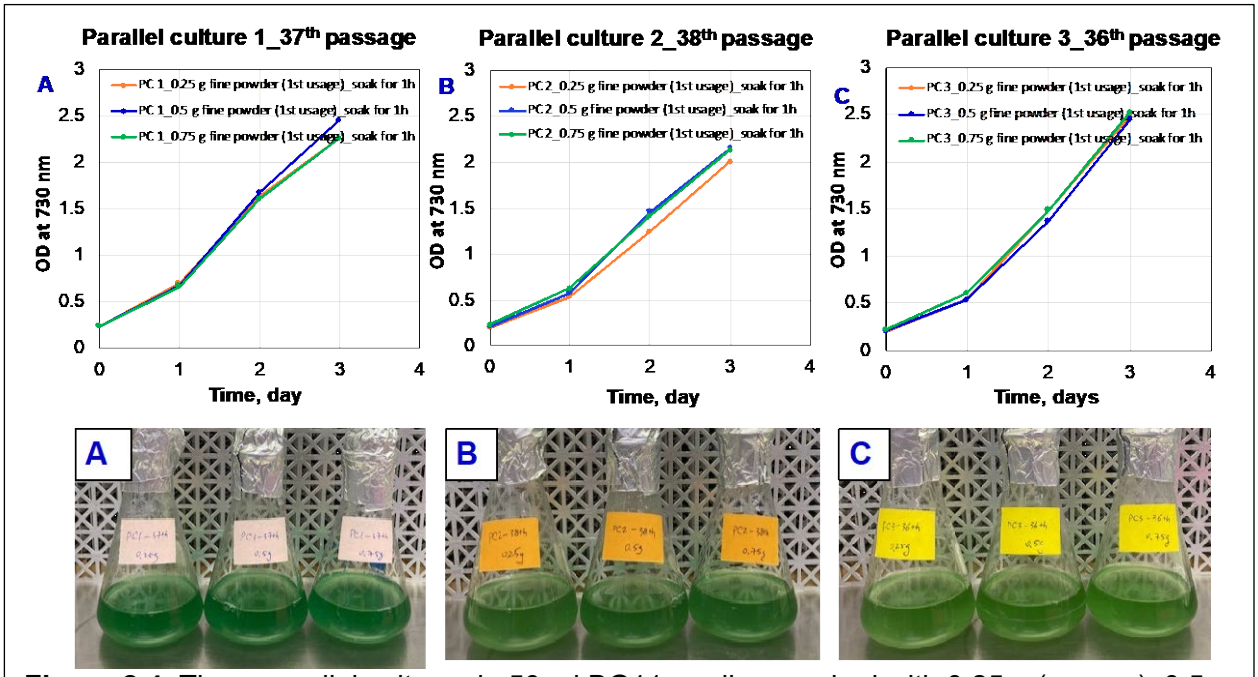


Figure 2.4. Three parallel cultures in 50 ml BG11 medium soaked with 0.25 g (orange), 0.5 g (blue) or 0.75 g (green) toxic methacrylate polymer for 60 min prior to inoculation of the cultures. Cultures were supplemented with 20 mM bicarbonate each day. The photos show the cultures 24 h after inoculation.

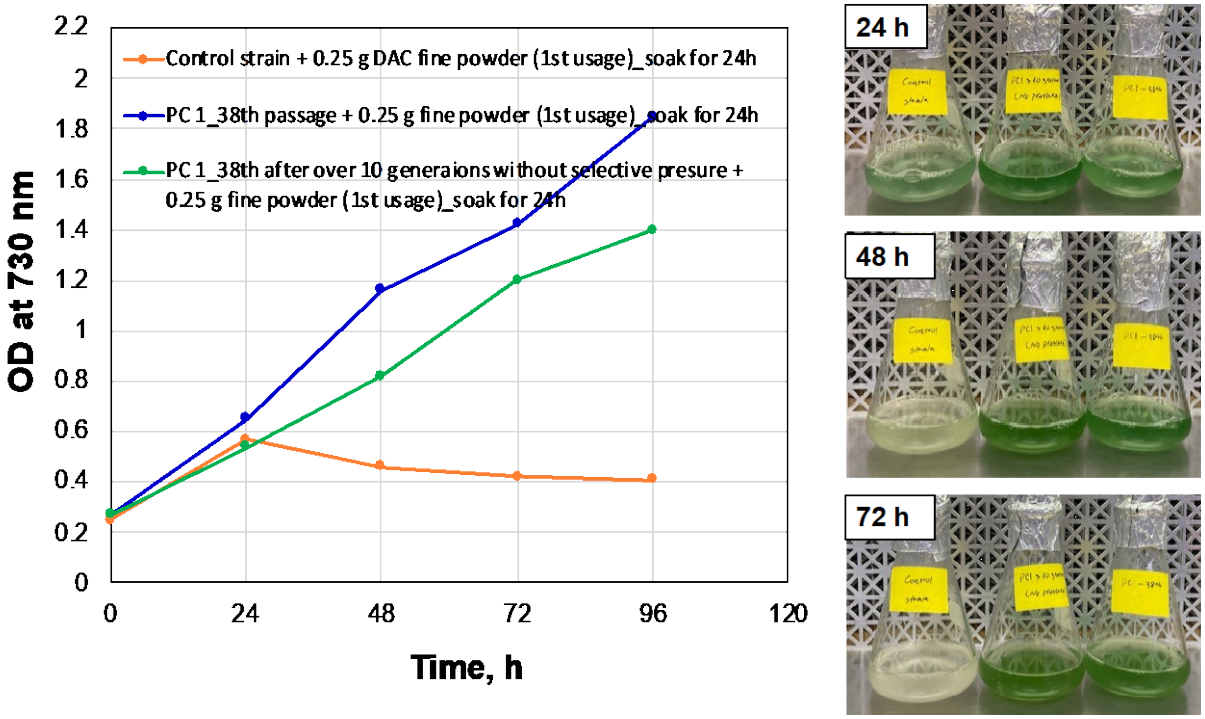


Figure 2.5. Tolerance of an evolved *Synechocystis* strain (right flasks) vs. the unevolved parent strain (left flask) to methacrylate polymer. Cultures were supplemented with 20 mM bicarbonate daily.

Currently, the evolved *Synechocystis* strains from three parallel cultures along with the parent strain have been frozen and are now being revived on BG11 plates for preparation of cultures for DNA extraction and sequencing to determine common genetic changes in the three parallel cultures.

A4. AUDACity 1g cultivation

Milestone 2.2.3: Lab-scale demonstration of AUDACity *1g* system for cyanobacterial cultivation meeting technical targets in Milestone 2.2.1 (M24).

After evaluating the performance of an Excellion belt at various cycle times in the updated 1 g system, *Synechocystis* was grown in 12 L of BG11 medium with a 4-h cycle. The temperature of the culture was 32 ºC. The culture was circulated well, so that *Synechocystis* cells did not settle.

The growth of *Synechocystis* cells over a course of two weeks in the 1 g system is shown in **Figure 2.6.** In this figure, the OD of the culture shown was an average of four data points that were taken from four different places in the 12 L culture. *Synechocystis* cells grew fine in the

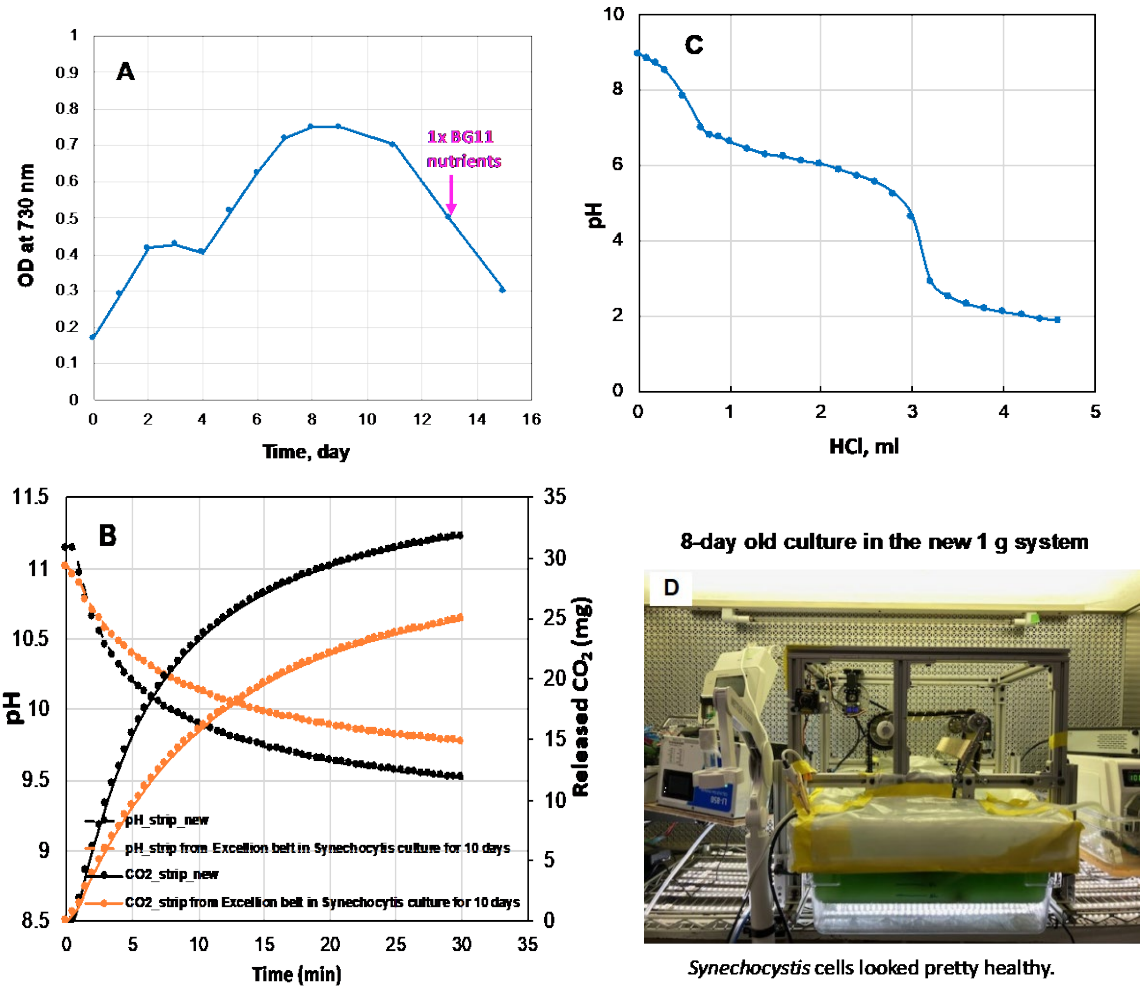


Figure 2.6. A) *Synechocystis* was grown in 12 L of BG11 medium with the Excellion belt on a 4-h cycle; **B)** CO_2 release kinetics from a new Excellion strip and one that had been used for 10 days; **C)** Titration of a sample from the 1 g *Synechocystis* culture with 0.1 M HCl; **D)** 8-day old *Synechocystis* culture in the 1 g system.

first two days, then stopped growing for two days and then started growing again. The growth trend of *Synechocystis* in the new 1 g system is very similar to that in the old 1 g system. The reason for irregular growth is not clear, but it could be caused by deficiency of phosphate or inorganic carbon. Nitrate deficiency is unlikely as the cells looked healthy and retained phycobilisomes.

To determine what may have caused the poor growth of *Synechocystis* in the 1 g system, CO_2 release and kinetics from one of strips on the Excellion belt were tested and compared to that of a new strip. **Figure 2.6B** showed that the Excellion strip on the belt still had close to 80% of the initial CO_2 release capacity. In addition, the bicarbonate concentration in the medium on day 12 of the 1 g system was estimated by a titration with HCl. 25 ml *Synechocystis* culture from day 12 was centrifuged and the supernatant was collected for titration with 0.1 M HCl. The titration of the *Synechocystis* culture with HCl confirmed that there was sufficient bicarbonate (~10 mM) in the medium (**Figure 2.6C**). The lagging growth in the 1g system most likely was caused by BG11 nutrients binding to the Excellion belt and thus depriving the culture of these nutrients.

Unfortunately, the Excellion belt broke a couple of times during the growth in the first week and fell into the medium; moreover, the belt broke rather frequently after day 10 due to some clips on the strip coming loose. The Excellion belt was taken off after day 11, some clips were replaced and the belt was put back on day 13. The *Synechocystis* culture was run without the Excellion belt during this period of time.

We are now gearing up for 1g system experiments with a new belt design and different polymer (A501).

A5. Phycocyanin

We used milk fat globule membranes (MFGM) to encapsulate phycocyanin in presence of dimethylformamide (to resuspend 10 mg phycocyanin) and cyclohexane (to dissolve 1 mg milk fat globule membranes). The dried pellet, after encapsulation, was resuspended in 50 mM sodium phosphate buffer (pH 7.5) to separate largely unencapsulated phycocyanin (which dissolves in water) along with smaller encapsulated particles that do not pellet, from larger particles of phycocyanin that are readily pelleted, including larger encapsulated particles.

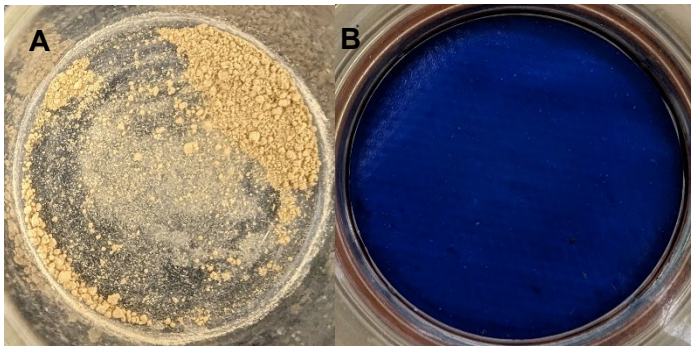


Figure 2.7. A. Phycocyanin denatured by incubating at 250 ^{o}C for 24 hours and used as a negative control. B. Phycocyanin particles on a 0.2 µm filter membrane after washing away free (unencapsulated) phycocyanin.

From our previous experience, we know that the phycocyanin pellet (after encapsulation) may contain empty non-fluorescent particles that do not contain phycocyanin. Thus, to separate fluorescent particles of a specific size range, we used fluorescence-activated cell sorting and selected the fluorescent particles of the size range: 1 – 10 µm. Before cell sorting, we filtered the pellet to obtain particles of a specific size range. To remove any particles smaller than 0.2 µm (**Figure 2.7.B**), we filtered the pellet using 0.2 µm membrane filters followed by filtering the filtrate through a 35-µm filter to remove any larger aggregates. The resulting particles of size range 0.2 – 35 µm were sorted using flow cytometry where denatured phycocyanin (baked at 250 ^{o}C for 24 hours) was used as a negative control (**Figure 2.7.A**). 27% of the total parent phycocyanin particles were sorted by the flow cytometer and these particles were used in our thermostability study.

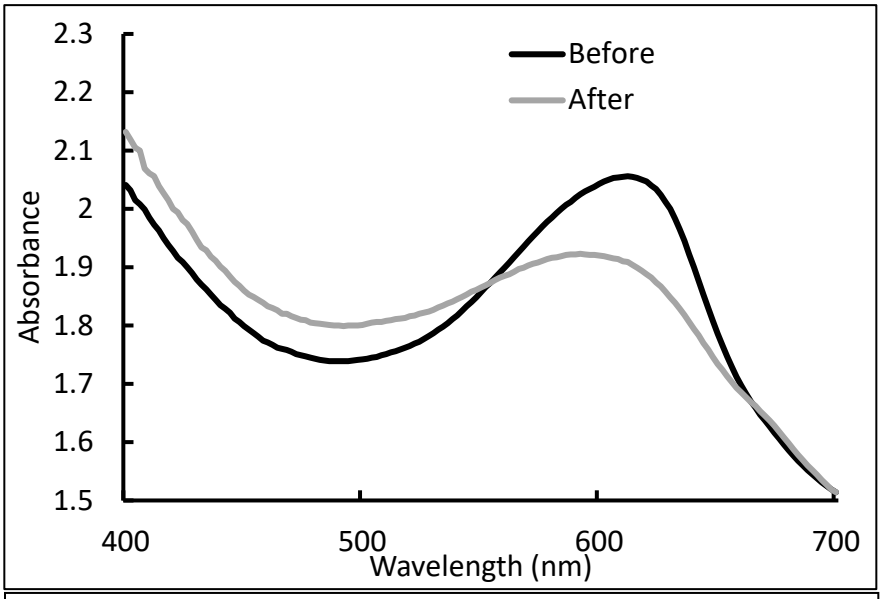


Figure 2.8. Thermostability of *Synechocystis* phycocyanin encapsulated in milk fat globule membranes. The flow-sorted phycocyanin was incubated in 50 mM sodium phosphate buffer (pH 7.5) at 100 °C for 10 minutes.

The sorted phycocyanin particles were subjected to heat treatment (100 °C for 10 minutes). Similar to our previous report, after the heat treatment, we observed about a 25% loss of color (**Figure 2.8**), and the pellet was still intensely blue.

Here we used the fluorescence-activated cell sorting system to sort fluorescent particles in the mixture and select fluorescent particles of a specific size range. The cell sorting we performed was not specifically optimized for the phycocyanin; thus we suspect some of the remaining free phycocyanin may have been selected by the cell sorter. We further plan to optimize the cell sorting system by modulating the pressure of the flow cytometer, concentration of the sample, flow rate of the sample as well as making sure that we filter out any free phycocyanin from the encapsulated sample before subjecting it to the flow cytometer. We will also perform time-dependent thermostability analysis of the sorted encapsulated phycocyanin after we optimize the particle sorting system.

Again, this shows that the encapsulated phycocyanin was largely thermostable: the phycocyanin particles that are covered by lipid globules do not directly interact with water, thus retaining its color upon boiling.

3. Explanation of Variance:

<u>Biocompatibility</u>

- We have not sequenced the evolved *Synechocystis* strains in this quarter. We did many tests this quarter to confirm that the evolved *Synechocystis* strains actually had

developed permanent tolerance to the methacrylate polymer before we ended the strain evolution experiment. We will sequence the evolved *Synechocystis* strains next quarter to determine common genetic changes in the three parallel cultures.

- We have not used the evolved strains with improved methacrylate tolerance to test biocompatibility of newly synthesized polymers yet, but will test them in the next quarter.

AUDACity 1g cultivation

Synechocystis cultivation in the updated 1 g system was not successful due to the frequent crashes of the Excellion belt. A new 1 g system was installed in the laminar hood recently updated with new plastic roller chain that holds the sorbent on the gears tightly. In addition, the new 1 g system was updated with 16 “pixie stick” packets holding 16 g sorbent A501. Currently, we are evaluating the performance of the newly updated 1 g system and then will cultivate *Synechocystis* in the updated 1 g system.

4. Plans for Next Quarter:

Biocompatibility

- Continue to test the biocompatibility of polystyrene-based polymers with *Synechocystis* cultures; from this the best-performing one will be selected for large-scale cultivation.
- Sequence the evolved *Synechocystis* strains that are tolerant to the methacrylate polymer to determine common genetic changes in the three parallel cultures.

AUDACity 1g cultivation

We will begin cultivation trials using with the most promising powdered commercial polymer (A501) contained within the mesh pixie stick design. Furthermore, by extending the belt to increase drying time, we plan to be able to meet the growth targets laid out in Milestone 2.2.1 under continuous and diurnal light conditions.

Phycocyanin

We plan to optimize the flow cytometry system to sort the phycocyanin particles in the desired range (0.1 – 10 μm). In addition, we plan to explore how the pH of the protein extraction buffer affects the encapsulation efficiency and the interaction between protein and milk fat globule membranes. We will extract phycocyanin in sodium phosphate buffer of different pH and encapsulate thus extracted protein in milk fat globule membranes. With this information we will be able to understand how the protein surface charge affects its interaction with milk fat globule membranes and improve the thermostability of the encapsulated phycocyanin. These optimization approaches will help us understand the interaction between protein and milk fat globule membranes, increase the encapsulation efficiency, and optimize particle sorting.

Task 3: Design, develop and evaluate AUDACity systems (M4–36)

1. Planned Activities:

- Continue testing and iterating mesh framework belt design. Implement new “pixie stick” design of conveyor system for powdered/beaded sorbents (see Figure 3.2 in the FY2022-Q4 report).
- Measure A501 sorbent water uptake and drying when uncoated or coated with Starbrite.

- Collect and analyze data from the AUDACity 1g system in the laminar flow hood.
- Continue design refinement and testing of a second AUDACity 1g system for abiotic evaluation including operation for four weeks.
- Complete the 100g system design and start building.

Deliverables/Milestones:

Deliverable 3.3.1: AUDACity 1g system developed and evaluated outdoors for four weeks correlating CO_2 sorption rates with wind speed, temperature, and relative humidity (M24).

Deliverable 3.4.1: Develop AUDACity 100g systems; lab-tested operation of the control system and demonstrated CO_2 delivery into abiotic medium will meet demand (M24).

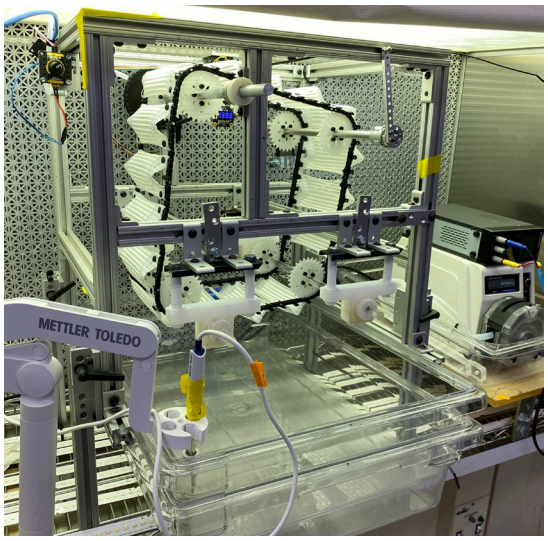


Figure 3.1. Updated 1 g laminar flow system with a belt of 16 “pixie stick” style packets containing 1 g of A501 sorbent resin each.

2. Accomplishments:

AUDACity 1g A501 belt

The 1g Laminar Flow unit was updated to include the four-axle geometry used in the 1g Rooftop system (**Figure 3.1**). This allowed for a longer belt resulting in longer allowable dry (adsorption) time while still maintaining the same wet (desorption) time. Additionally, a new belt was installed on the 1g system composed of the new plastic roller chain with attachment points and elongated tubular mesh packets holding beaded sorbent resin A501 we call “pixie stick” packets. This replaced the old solid-form Excellion sheet belt that used an unreliable Technic chain. Several packets were attached using nylon eyebolts (**Figure 3.2**), allowing researchers to easily remove individual packets at specific positions along the cycle. By evaluating packets at critical points in the adsorption-desorption cycle, we can test performance, moisture content, or collect other important data to assist in determining ideal cycle times and operating parameters. The system was also updated with a new tensioning axle to reduce potential for the chain slipping off its gears.

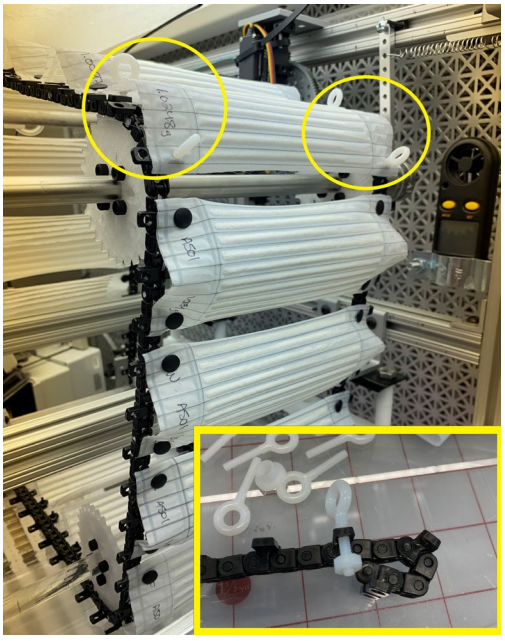


Figure 3.2. Sorbent packets with easily removable nylon bolts. Bottom right shows the packet attachment points without packet.

Sorbent water uptake and drying

Last quarter we began transitioning water uptake tests from manually weighing samples at various intervals using a water blotting technique to hanging packets of sorbent on load cells for continuous data recording to remove variability in the amount of water removed by blotting leading to far more reproducible results.

Figure 3.3 shows the change in mass of the sorbent after being soaked in water for 1 hour and as it dries from several test runs of similar six-stick-configured packets. Nicely clustered data indicate the load cell tests are reproducible. This chart indicates a rather significant reduction of water uptake in A501 sorbent resin coated in Starbrite (SB) hydrophobic coating versus uncoated. This also resulted in a shorter drying time.

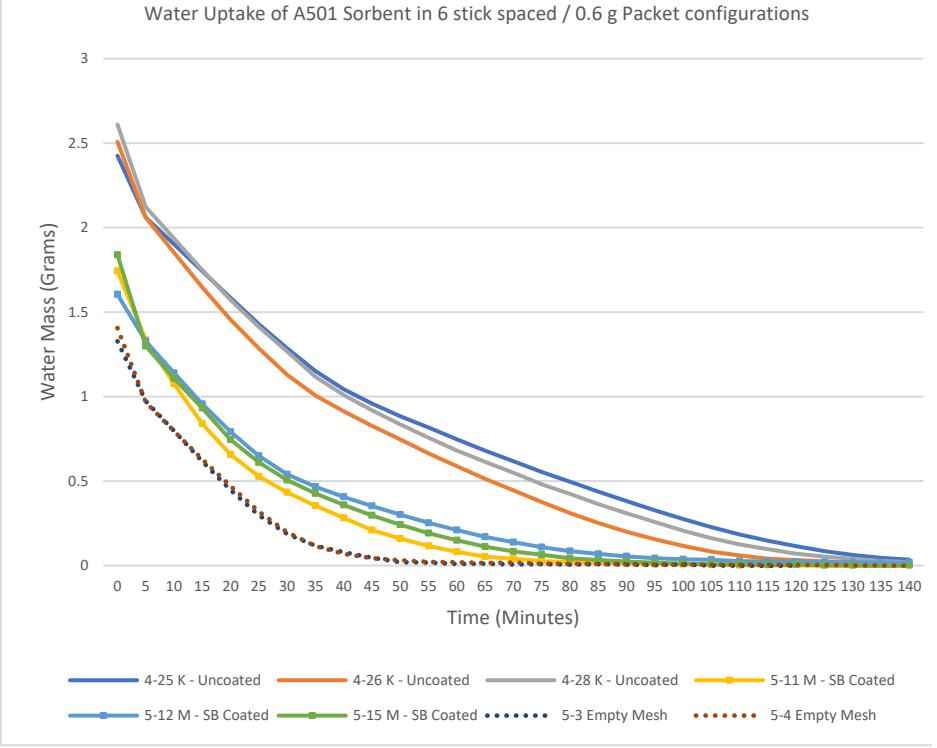


Figure 3.3. Water uptake in soak tests of A501-sorbent-loaded packets in a six-stick configuration. Packet K contained uncoated A501 sorbent resin totaling approximately 0.6 g. Packet M contained A501 with hydrophobic Starbrite coating (SB) totaling approximately 0.6 g. Empty Mesh packets were unfilled – containing no sorbent.

Temperature and relative humidity were recorded for every soak test (**Table 3.1**). Relative humidity has a major influence on drying time as seen on 4/25 with a relative humidity of 45% versus the 4/26 soak test the following day with relative humidity of 33%.

To further distinguish water uptake of the mesh versus sorbent, tests were run on a packet without sorbent (empty mesh). Surprisingly, the empty packet was found to take up a significant amount of water, i.e., over 50% of the water levels seen in the loaded, uncoated packets. With this information, packets were reconfigured into a non-spaced format, reducing the amount of mesh to sorbent ratio. New 10-stick packets were created with each stick loaded with 0.1 g A501 (**Figure 3.4**). This new

configuration has become standard for not only the 1 g system belts, but also for new wind tunnel tests determining performance capacity of sorbent resins.

Table 3.1. Temperature and relative humidity for every packet soak test to determine if lab conditions contribute to differences in water uptake.

Packet	K Bag			M Bag			Empty Mesh	
Config	6 sticks			6 sticks			6 sticks	
S/NS	spaced			spaced			spaced	
Density	.1g each stick			.1g each stick			n/a	
Coat	uncoated			Starbrite coated sorbent			uncoated	
Sorbent.	.6185 g A501			.6196 g A501			0 sorbent	
Date	4/25/2023	4/26/2023	4/28/2023	5/11/2023	5/12/2023	5/15/2023	5/3/2023	5/4/2023
Dry Temp	20C	21.4C	19.7C	21.3C	19.6C	21.2C	23.1C	23.1C
Dry RH	46%	33%	41%	40%	46%	43%	42%	31%
Wet Temp	21.5C	22.2C	21.1C	22.4C	23.2C	21.6C	22.3C	21.3C
Wet RH	41%	30%	35%	37%	40%	41%	43%	42%

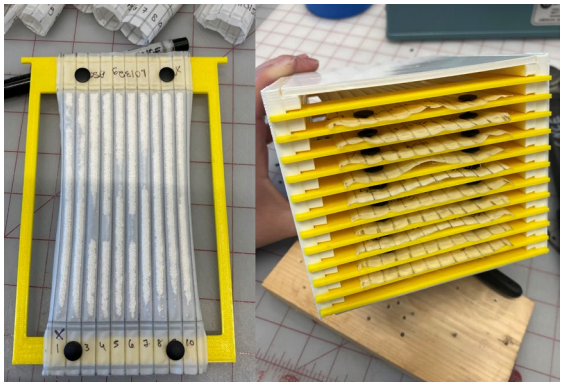


Figure 3.4. (*left*) Ten-stick, 0.1 g per stick distribution packet containing A501 sorbent on frame for wind tunnel testing. (*right*) 10 framed packets in 3D printed jig ready for wind tunnel test.

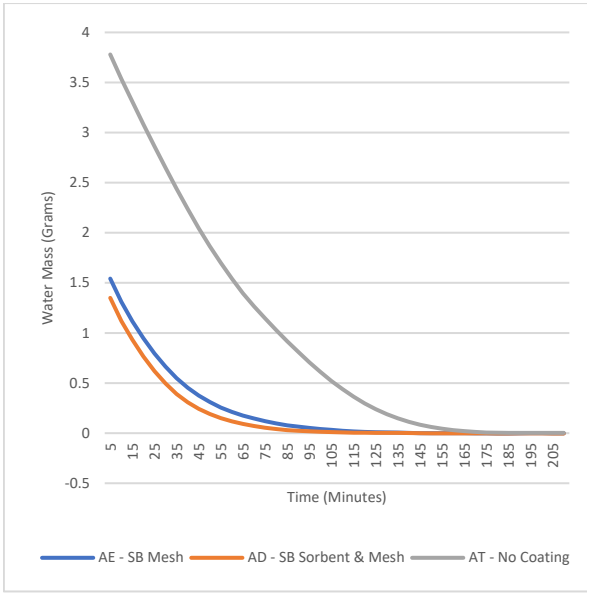


Figure 3.5. Water uptake in soak tests of A501-sorbent-loaded packets in a ten-stick configuration. Each packet contained 1 g of A501 sorbent resin.

The load cell was also used to test water uptake and drying in the new 10 stick / 1 g configuration comparing an uncoated packet to a packet with mesh coated in Starbrite with uncoated A501 resin to a mesh coated in Starbrite that also contained Starbrite coated A501 resin. The results of four tests with the load cell averaged are shown in **Figure 3.5**, which shows a significant water uptake reduction when using Starbrite-coated mesh packets but relatively little difference between coated versus uncoated sorbent within the coated mesh packets. Further tests will be run to determine changes to the water uptake in the coated mesh packets over time after repeated use.

Mesh Water Uptake and Percent Openness Tests

As shown in Figure 3.3 with empty mesh packet load cell results, tests have been conducted to determine optimal mesh materials. **Figure 3.6** shows results of mesh sample water uptake tests. The first mesh, MM40 is a polyester material with a pore size of 40 microns, while sample A25 is a nylon material and has 25-micron pores. A25 has been the standardized mesh for all pixie stick packets thus far. Another sample of A25 mesh was tested after treating with Starbrite hydrophobic coating. MM40 has a much lower water uptake than A25; however, the Starbrite coating reduced water uptake in A25 significantly. This matches data trends in Figures 3.3 and 3.5.

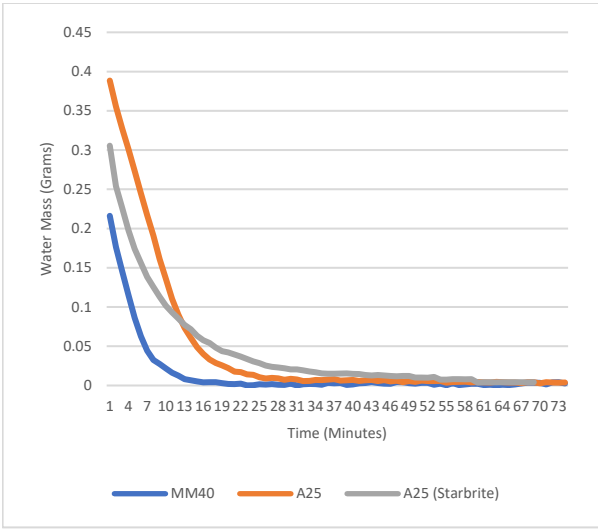


Figure 3.6. Water uptake in soak tests of blank mesh sheets. Samples include MM40 (40-micron pore size polyester with 25% openness), and A25 (25-micron pore size nylon with 8% openness) with and without Starbrite coating.

A critical parameter for ensuring rapid CO_2 delivery is the mesh pore size and the ratio of open pores (empty space) versus closed cloth of the mesh, also known as percent openness. To determine percent openness of samples MM40, A25 and A25 with

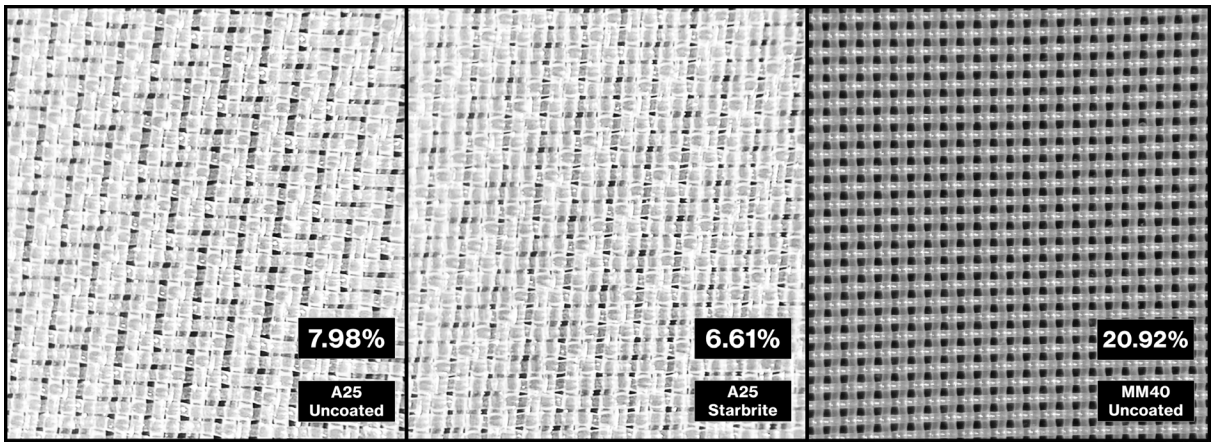


Figure 3.7. Mesh samples of A25 before and after Starbrite Coating and uncoated MM40. Percent openness is calculated as the ratio of open pore to closed fabric of the mesh.

Starbrite coating, photos were taken at high magnification and then processed to create high contrast black and white images with high resolution. Using photoshop (and repeated with similar results using ImageJ), images' open pores were selected with a color selection tool and the pixel proportions in histogram determined vs total pixels. **Figure 3.7** shows determined percent openness of the samples. MM40 had a much higher percent open than either coated or uncoated A25. It is also important to note that adding the Starbrite hydrophobic coating may reduce percent openness. Typically, a higher percent openness resulted in faster kinetics: sorbent could adsorb/desorb CO_2 and dry quicker. A shorter drying/adsorb period means a shorter full cycle time necessary in the system and more cycles possible per day. More cycles will lead to less sorbent, packets, and length of belt necessary in the system. Recent soak tests

indicate a ~90 minute drying time with a 30 minute desorb/wet time. As we find more optimal mesh samples, we will modify the system design to accommodate shorter cycle times.

Design of the 100g system

The 100g system design was updated based on results of packet testing and sorbent performance. Consideration was given to sorbent delivery and cycle times when calculating the necessary number of “pixie stick” packets, length of belt, and overall geometry.

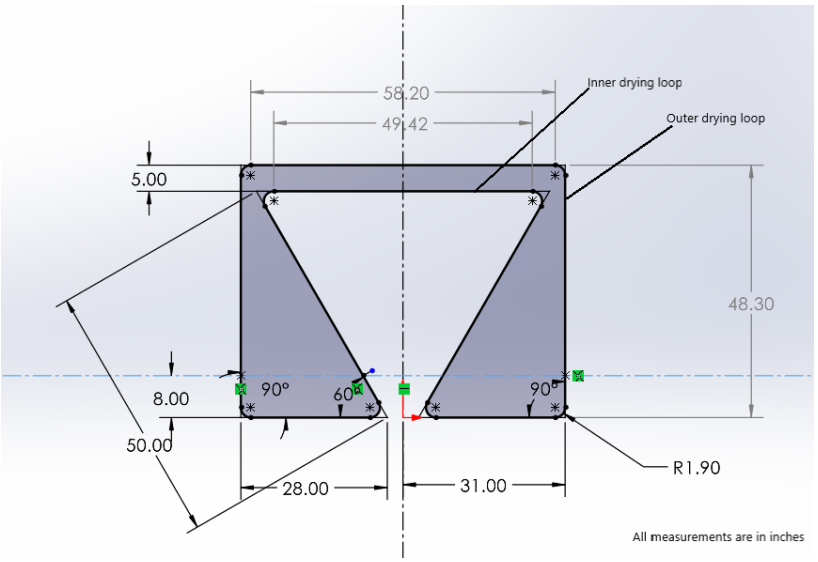


Figure 3.8. Preliminary belt path calculation.

Figure 3.8 illustrates a 100g design that incorporates a dual carbon capture cycle per belt rotation, aimed at reducing power usage to rotate the belt and reducing shading over the pond surface that would otherwise slow cyanobacterial growth. By adopting this design, not only will the overall power usage be reduced, but it will also reduce the size of the entire system. The design assumes that each packet needs approximately 30 minutes for desorbing CO_2 in the liquid phase and 90 minutes for drying and adsorbing CO_2, resulting in a wet travel length vs. dry travel length ratio of 1:3. The total length of the belt is 356.92 inches (Drying Loop = 273.28 in, Wet Loop = 83.63 in). The design of the outer and inner drying loops has a length difference of 8.64 inches (Outer Drying Loop = 140.96 in, Inner Drying Loop = 132.32 in), which is only about 6%.

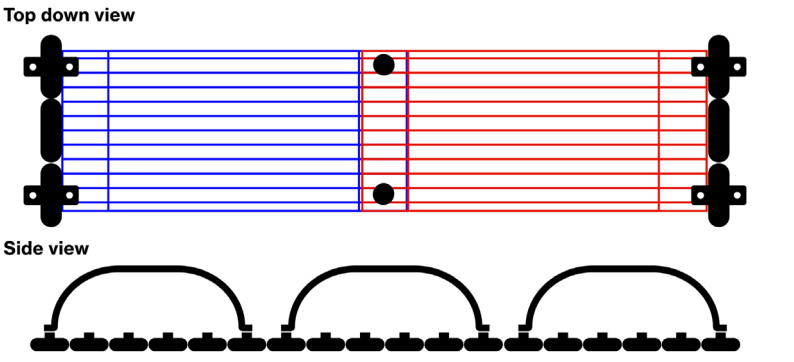


Figure 3.9. Packet placement on belt.

The 100g design was also updated to expand the belt be two packets wide. Unlike the single-row arrangement of packets in the 1g system, the decision was made to position two packets side by side, creating two rows along the belt in the 100g system (**Figure 3.9**). This choice was made considering that the 100g system will consist of approximately 250-300 packets.

Sorbent drying and loading in the wind tunnel

The wind tunnel provides a systematic and controlled environment for measuring sorbent drying and loading after specific unloading conditions. During this quarter, tests were performed in the wind tunnel to evaluate the effect of different Na_2CO_3 buffer solutions on A501 sorbent desorption and the effect of wind velocity on A501 sorbent drying and loading. The wind tunnel

was also used to compare sorbent drying and loading when the sorbent powder was packed into a single pouch polyester mesh packet compare the “pixie-stick” form factor composed of elongated mesh tubes. The experimental method and results are described in the following subsections.

Test method

The A501 sorbent was ion-exchanged to bicarbonate following protocols described in the FY23-Q2 report (under Task 1). The sorbent was inserted in a polyester mesh bag and exposed to ambient air for 24 hours to equilibrate with atmospheric CO_2 and H_2O vapor partial pressures. Once equilibrated the sample was submerged in an aqueous solution to desorb CO_2 before the wind tunnel absorption experiment. The first set of experiments evaluated the impact of the solution buffer concentration on the amount of CO_2 absorbed within the wind tunnel. Desorption was performed with three different aqueous solutions: 1) deionized (DI) water only, and buffer solutions containing either 2) 10 mM Na_2CO_3 or 3) 100 mM Na_2CO_3. The sample was submerged under these conditions for 1 hour to ensure equilibrium was reached for each desorption condition given prior data showing desorption was complete within about 30 minutes. The sorbent was then rinsed with DI water for conditions 2 and 3, and for all conditions the sorbent was left for 5 minutes on the sample holder to remove excess water from the surface so it would not get blown into the wind tunnel. The sorbent packet frame was then inserted on the wind tunnel chamber with the set wind velocity; in all cases the humidity was set to be dry between 3 and 5 ppt H_2O and 25 °C.

The second set of experiments evaluated CO_2 absorption in the wind tunnel under different wind velocities: 0.7, 2.5 and 5 m/s utilizing the pixie-stick form factor. The third set of experiments evaluated differences in drying and CO_2 loading kinetics for the sorbent when contained within the pixie-stick mesh form factor versus the sorbent packed within a singular polyester mesh bag at 5 m/s wind velocity.

Results

Figure 3.10 show the results of the first set of experiments. All three scenarios begin with a phase lasting 0.4 h with a stagnant CO_2 concentration when there is a considerable amount of liquid water being released by the sorbent. During this time the sorbent’s pores were likely still filled with liquid water, which significantly slowed CO_2 from releasing into the gas phase or CO_2 from the air absorbing to the sorbent. After 0.4 h, the experiments utilizing DI H_2O only and 10 mM Na_2CO_3 solution start a phase where CO_2 is unexpectedly desorbed. The sample desorbed 154 and 50 µmoles CO_2/g when immersed in DI H_2O only and 10 mM Na_2CO_3 solution, respectively. This shows the sorbent was not fully unloaded with CO_2 at the start of the absorption experiment because neither solution had sufficient capacity to hold all the CO_2 being desorbed. The buffer concentration has shown to play an important role in creating a driving force for CO2 to desorb down a concentration gradient into solution. In the next experiment the sorbent was added to a 100 mM Na_2CO_3 solution during an attempt to fully desorb CO_2 from the sorbent. This experiment showed no CO_2 desorption after the initial 0.4 h of sorbent drying time. The total absorption capacity in each experiment was 670, 289, 121 µmoles CO_2/g respective to the 100 mM Na_2CO_3, 10 mM Na_2CO_3 and DI water only scenarios. The experiment with the 100 mM Na_2CO_3 buffer solution concentration leading to 670 µmoles CO_2/g absorption is in good agreement with the measured IEC of this material (e.g., 1000 µmoles CO_2/g, which is half the measured IEC of the material of 2.0 mmol charge per g) and is thus representative of the sorbent’s total capacity at the given temperature (25.8 °C) and relative humidity (15-20%)

established in the wind tunnel, and realizes a significant fraction of the potential CO_2 absorption capacity of the sorbent.

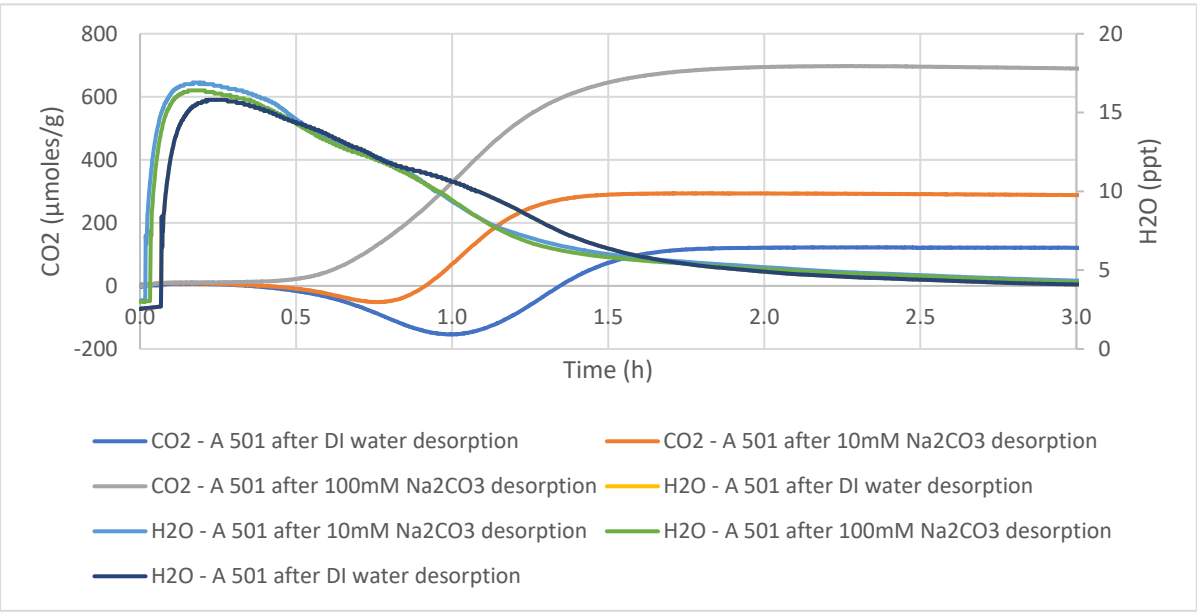


Figure 3.10. CO_2 absorption of A501 in the wind tunnel at 5 m/s wind velocity as a function of the buffer concentration of Na_2CO_3 used to desorb the sorbent.

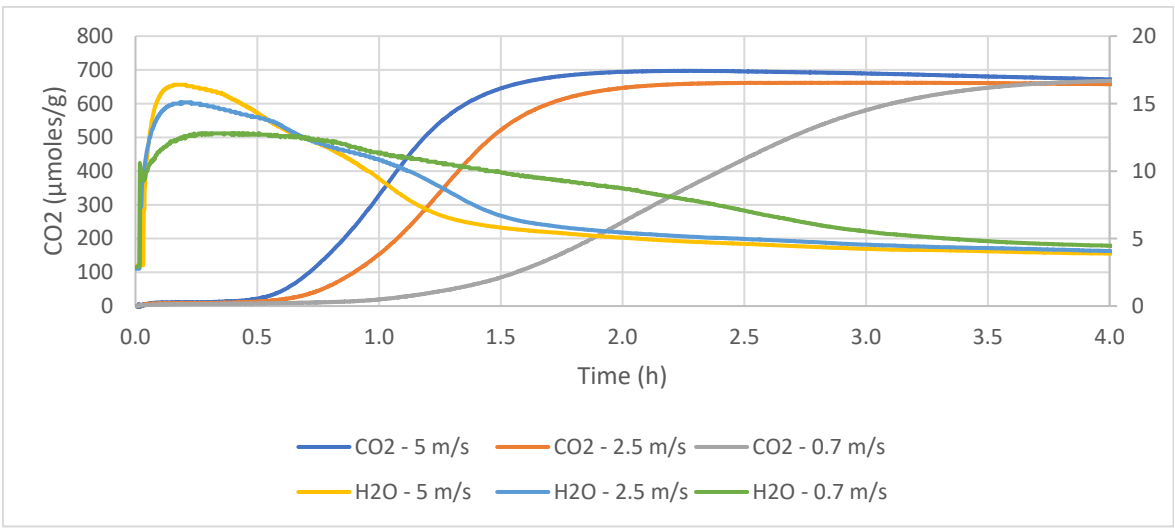


Figure 3.11. CO_2 absorption of A501 in the wind tunnel at 0.7, 2.5 and 5 m/s wind velocity after desorbing in aqueous media with 100 mM Na_2CO_3.

Figure 3.11 shows CO_2 absorption as a function of three different wind velocities: 0.7, 2.5 and 5 m/s. In all these experimental scenarios the sorbent was submerged in a 100 mM Na_2CO_3 solution for 1 hour, which was found previously to fully unload the sorbent. CO_2 absorption is observed starting after 0.4, 0.6 and 0.9 hours at 5, 2.5 and 0.7 m/s wind velocity, respectively, indicating longer time to dry the sorbent sufficiently for CO_2 absorption to begin. In all scenarios the sorbent was able to absorb 670 µmoles CO_2/g as expected from the previous set of experiments. As expected, the wind velocity plays an important role on the absorption kinetics.

The sorbent took 2, 2.5 and 3.8 hours to fully load at 5, 2.5 and 0.7 m/s, respectively. When considering the primary CO_2 loading phase after the initial drying phase at each wind velocity, it took 0.8, 0.9 and 1.9 hours to absorb about 500 µmoles CO_2/g (71 % of the total capacity) for 5, 2.5 and 0.7 m/s respectively, or about twice as long at 0.7 m/s vs 5 m/s. 0.7 m/s is close to the wind velocities in the laminar flow hood, suggesting that about 4 hours are needed to fully dry and load the sample under those conditions. In outdoor air, we expect average wind speeds closer to 2.5 to 5 m/s (5-10 mph) indicating we should expect drying/loading times closer to 2 or 2.5 hours at room temperature, though higher outdoor temperatures and direct sunlight might significantly decrease drying times.

Figure 3.12 compares sorbent drying and loading in the wind tunnel when packed into elongated thin mesh tubes (e.g., pixie stick; Figure 3.1 and **Figure 3.13**), which allows a better

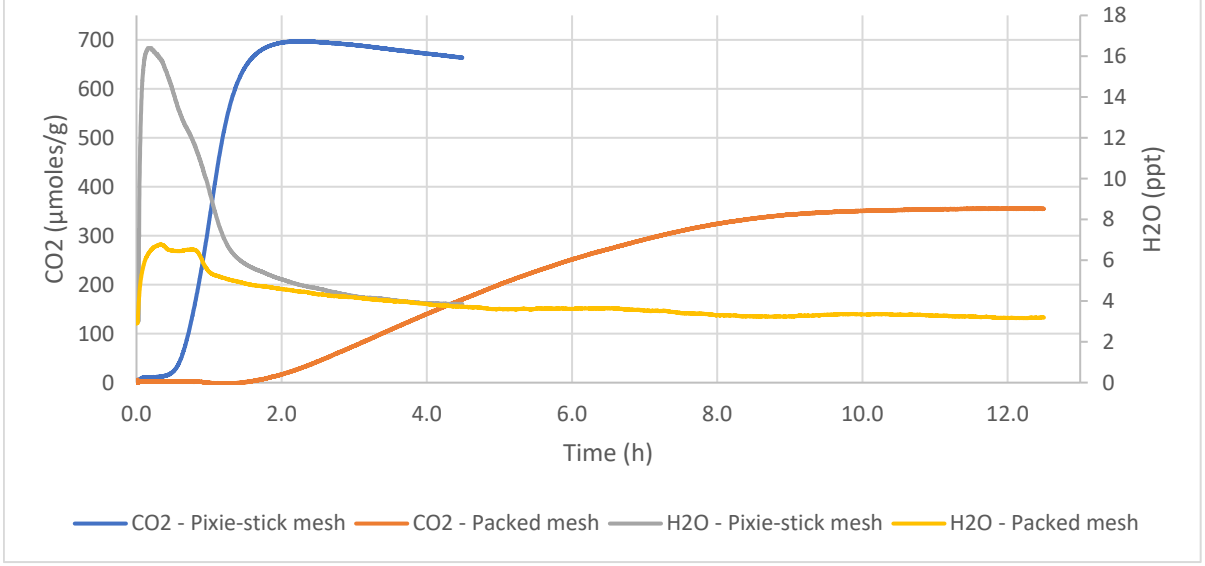


Figure 3.12. CO_2 absorption of A501 in the wind tunnel as a function of form factor (pixie-stick vs packed mesh) at 5 m/s wind velocity after desorbing in a 100 mM Na_2CO_3 solution.

distribution of the sorbent beads in order to allow more air to contact to the sorbent and faster drying times compared to when the sorbent is packed in two larger polyester mesh bags. Both experiments were conducted at 5 m/s wind velocity after desorbing the sorbent in a 100 mM Na_2CO_3 buffer solution for 1 hour. The pixie stick form factor has shown remarkably improved loading kinetics when compared to the packed sorbent in a polyester mesh bag. The packed mesh took 1.8 h to dry before absorption begins while the pixie stick form factor only took 0.4 h. The packed mesh sorbent appeared unable to reach the same capacity as when loaded in the pixie stick form factor, taking 12 hours to reach 355 µmoles CO_2/g while the pixie stick form factor only took 2 hours to reach 670

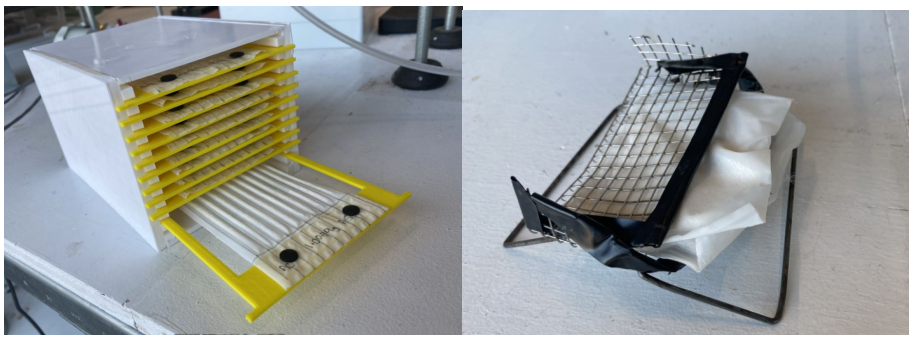


Figure 3.13. Pixie-stick form factor on rectangular rack versus packed mesh bag images.

μmoles CO_2/g. This represents a 6-fold faster total experimental time improvement. We hypothesize that the 1.9-fold apparent increase in capacity is due either 1) to a leak in the wind tunnel or 2) tight sorbent packing in the packed mesh that renders some of the sorbent unable to effectively absorb CO_2. The suspected wind tunnel leak can be seen Figure 3.12 in the pixie stick form factor as a decline in CO_2 absorbed (or CO_2 being released) from hours 2 to ~5, which if extrapolated over many more hours that it takes the packed mesh sorbent to load would significantly reduce the maximum absorbed CO_2 capacity. This hypothesis will be tested next quarter by running a wind tunnel experiment without a sorbent but instead adding a known quantity of CO_2 and monitoring the rate of CO_2 decline at the same wind velocity in order to evaluate how significant the leak rate is.

3. Explanation of Variance:

Planned work on the second 1g "Rooftop" system was put on hold to concentrate on water uptake testing and optimizing the mesh packet design for scale up in the 100g system. The rooftop system has been located in the CNCE lab and used for testing and development of necessary modifications to the laminar flow 1g system that is located in the Vermaas laboratory. In this way, the system has been extremely helpful, though no abiotic testing or data have been acquired this quarter.

The 100g system design has advanced significantly but is not yet complete. Much of the delay has been due to uncertainty involving sorbent choice and delivery performance. As new sorbent resins have been identified and their performance characterized, the geometry and scale of the 100g system design has shifted dramatically. This quarter's concentration on A501 ion-exchange resin has led to a wealth of data and a more solidified design that we hope to have fully built on site by the end of the upcoming quarter.

4. Plans for Next Quarter:

- Continue testing of "pixie stick" design of conveyor system for powdered/beaded sorbents
- Continue measuring A501 sorbent water uptake and drying when uncoated or coated with Starbrite.
- Lifetime testing of Starbrite-coated mesh to determine degradation of coating
- Collect and analyze data from the AUDACity 1g system in the laminar flow hood.
- Continue design refinement and testing of a second AUDACity 1g system for abiotic evaluation including operation for four weeks.
- Complete the 100g system design and start building.
- Test wind tunnel leak rate by running an experiment without a sorbent

<u>Deliverables/Milestones:</u>

Deliverable 3.3.1: AUDACity 1g system developed and evaluated outdoors for four weeks correlating CO_2 sorption rates with wind speed, temperature, and relative humidity (M24).

Deliverable 3.4.1: Develop AUDACity 100g systems; lab-tested operation of the control system and demonstrated CO_2 delivery into abiotic medium will meet demand (M24).

Task 4: AUDACity outdoor cultivation trials at AzCATI (M7–33)

1. Planned Activities:

a. Work with EPA to approve TERA application to meet Deliverable 4.1.1.

b. Scale up the *Synechococcus* sp. PCC 11901 strain, including evaluating conditions indoor to reduce/eliminate contamination, in particular amoeba. We will troubleshoot the current seed train scale-up process looking to eliminate any possible source of amoeba contamination including persistence of cysts in cultivation systems indoors. This will include the utilization of better sterile filtration of media as well as plastic liners in panels and all-new spargers systems as it can be difficult to remove persistent contamination such as cysts from such systems.

Deliverables/Milestones for Next Quarter:
Deliverable 4.1.1: Complete application of TSCA (Toxic Substances Control Act) Experimental Release Application (TERA; M21).

2. Accomplishments:

Noting to report.

3. Explanation of Variance:

TERA. Significant progress was made drafting the TERA application. We plan to submit the TERA application by end of July, 2023. While we expect EPA to approve use of our genetically modified strain for our proposed AUDACity 100g outdoor cultivation within 30 days (end of August), which should be in time to support our fall trial, if needed, we will begin our outdoor cultivation trial with a non-GMO strain.

4. Plans for Next Quarter:

a. Work with EPA to approve TERA application to meet Deliverable 4.1.1.

b. Using new indoor protocols to scale up amoeba-free 11901 for an outdoor cultivation trial and compare against last year's performance.

Due to biocompatibility issues with the sorbent, outdoor trials (and related milestones and deliverables) have been delayed until Aug/Sept 2023, which will also delay **Deliverable 4.1.2:** Biomass, PC, laurate and methyl laurate quantity and quality validated (M27).

Task 5: Techno-Economic and Life Cycle Analyses (TEA / LCA) (M4–36)

1. Planned Activities:

There are no deliverables due this quarter. Work will focus on refining the pathways to meet end of project goals including the use of data from the 1 g scale testing to assess the environmental impact of the proposed biorefinery. A comparative techno-economic assessment of different polymers will aid in the selection process of a polymer to scale for the 100g system.

2. Accomplishments:

Key cost-driving parameters were updated with new experiment data. Results were updated and presented at Algal Biomass Biofuels and Bioproducts, Waikoloa Beach, HI. Adsorption and desorption rates were updated in the model based on new wind tunnel and abiotic desorption testing with polymer type A501. Average adsorption and desorption rates are required by the engineering process model to calculate polymer mass requirements. As seen in in quarterly

report FY23-Q1 (Reporting Period: 10/01/2022–12/31/2022), polymer mass requirements have a high impact on cost. Low rates increase polymer mass requirements increasing CO_2 delivery cost. Average adsorption and desorption rates are calculated from experimental data.

Data processing

To process the raw data, effective ion-exchange capacity must be assumed. This is defined as the amount of CO_2 that can be adsorbed and desorbed per cycle. Based on data from moisture swing tests and wind tunnel tests, the effective ion-exchange capacity was assumed to be 700 $\mu mol \cdot g^{-1}$. The processed data for adsorption is provided in **Figure 5.1**. In Figure 5.1 A, the polymer is completely desorbed when entering the drying cycle while in Figure 5.1 B it is still loaded with CO_2. A polymer may enter the drying cycle before completely desorbing its CO_2 for a number of reasons including but not limited to buffering capacity of the pond medium. Figure 5.1 A is likely more representative of implementation in an open raceway pond where CO_2 is actively consumed by algae and is therefore used in the subsequent TEA. Adsorption rates are dependent on relative humidity and concentration gradients. In Figure 5.1, the instantaneous adsorption rate decreases as the polymer becomes fully loaded with CO_2. Drying time decreases the average adsorption rate. Either a negative slope or a slope of zero signifies that the polymer is still drying. In both Figure 5.1 A and 5.1 B the polymer dries for approximately the same amount of time, 30–40 minutes. Design of a hydrophobic mesh is anticipated to reduce drying time, which could increase average adsorption rates. The current average adsorption rates in Figure 5.1 A and 5.1 B are 0.122 and 0.006 $\mu mol \cdot g^{-1} \cdot s^{-1}$, respectively. The processed data for desorption are provided in **Figure 5.2**. Desorption rates are dependent on concentration gradients, buffering capacity, algae growth rates and pH. It is suspected that concentation gradients limited desorption to ~350 $\mu mol \cdot g^{-1}$ during this test. The average desorption rate in Figure 5.2 is 2.368 $\mu mol \cdot g^{-1} \cdot s^{-1}$.

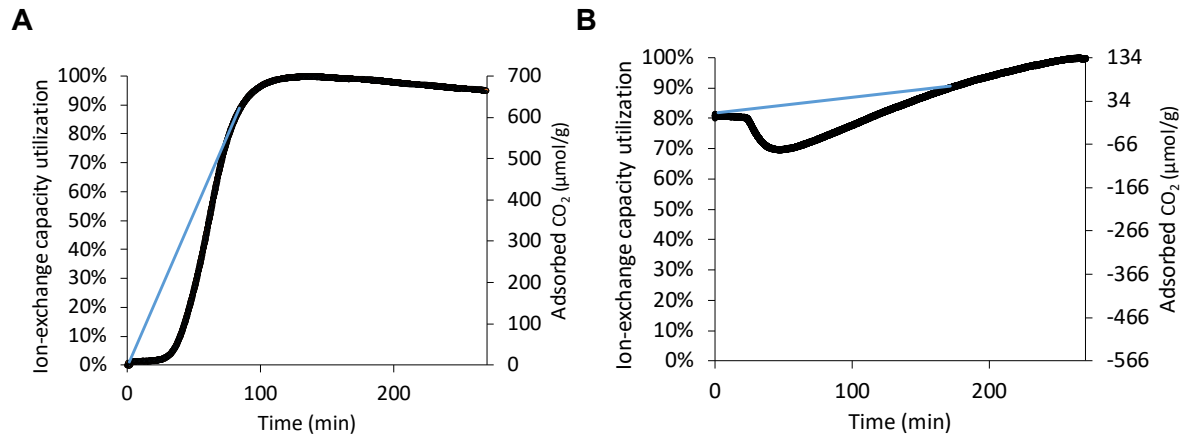


Figure 5.1: Wind tunnel adsorption data for polymer type A501 after: A) polymer desorbed in 100 mM K_2CO_3 solution for 1 hour and B) polymer desorbed in water for 1 hour. It was assumed that the ion-exchange capacity is 700 $\mu mol \cdot g^{-1}$. The blue line shows the average slope between the initial CO_2 loading and 90% loading (or 90% ion-exchange capacity utilization).

Techno-economic analysis and life cycle assessment

The average rates from experimental data were used in the engineering process model. It was assumed that the ion-exchange capacity was 700 µmol·g^{-1}, the adsorption rate was 0.122 µmol·g^{-1}·s^{-1} and the desorption rate was 2.368 µmol·g^{-1}·s^{-1}. They are the basis for the techno-economic results presented in **Figure 5.3**. The current scenario utilizes the average rates discussed above in Figure 5.1 A and Figure 5.2. The Future Scenario employs a higher ion-exchange capacity (1500 µmol·g^{-1}) and a higher adsorption rate (2 µmol·g^{-1}·s^{-1}), both of which are within reason for an optimized system. The grey bars represent state of technology CO_2 delivery systems for high purity point source capture as well as temperature swing direct air capture. These results were shared at Algal Biomass Biofuels and Bioproducts.

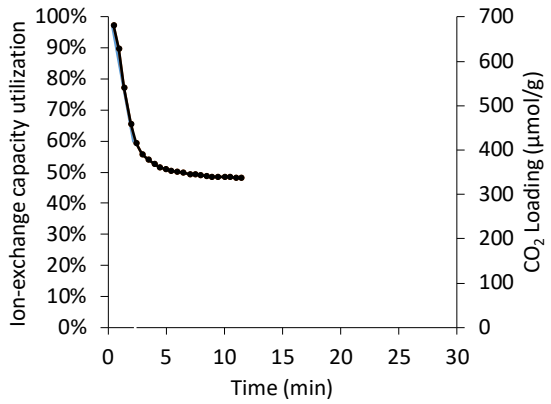


Figure 5.2: Desorption data for new polymer of type A501. It was assumed that the ion-exchange capacity is 700 µmol·g^{-1}. The blue line shows the average slope between the initial CO_2 loading and 60% loading (or 60% ion-exchange capacity utilization).

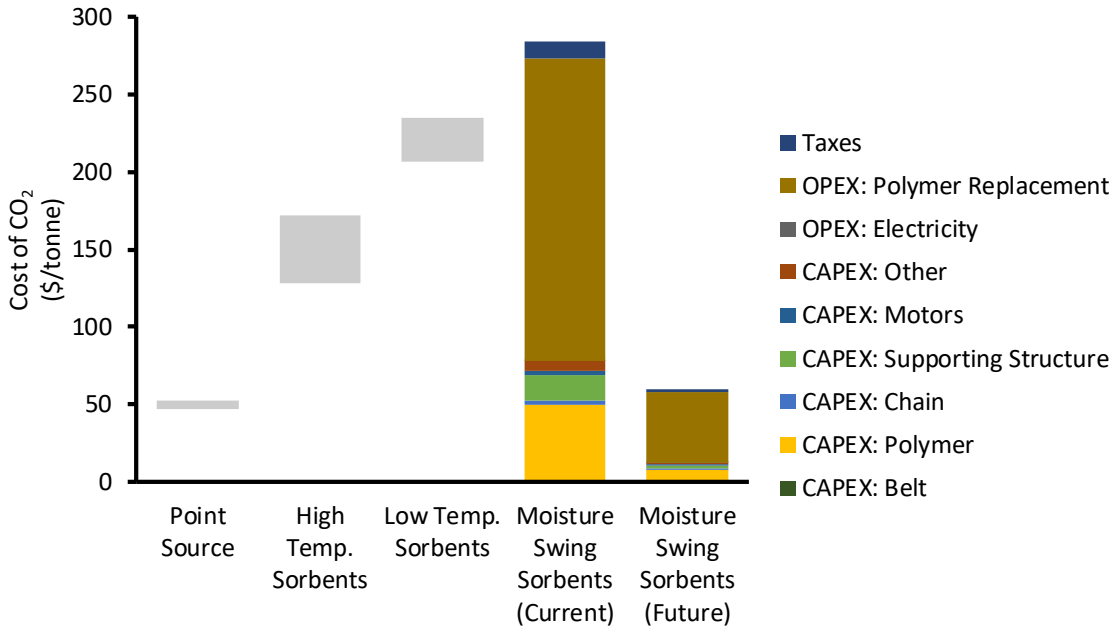


Figure 5.3: TEA of the AUDACity system.

3. Explanation of Variance: Data is still being collected for a proper comparative techno-economic assessment of different polymers. Data processing this quarter has laid the foundation for collecting data on other polymer types.

4. Plans for Next Quarter:

There are no deliverables due next quarter. Work will focus on refining the pathways to meet end of project goals including the use of data from the 1 g scale testing to assess the environmental impact of the proposed biorefinery. A comparative techno-economic assessment of different polymers will aid in the selection process of a polymer to scale for the 100g system.

References

Amel, A.; Gavish, N.; Zhu, L.; Dekel, D. R.; Hickner, M. A.; Ein-Eli, Y. Bicarbonate and Chloride Anion Transport in Anion Exchange Membranes. J Memb Sci 2016, 514, 125–134.

Section III. Issues, Risks and Mitigation

We can now synthesize polystyrene-based polymers with good CO_2 capacity and delivery kinetics that are biocompatible. Future work will try to reduce water uptake compared to commercially available polymers that will be used for initial outdoor cultivation trials at the 100 g CO_2 per day scale.

Section IV. Changes in Approach

Nothing to report.

Section V. Changes in Key Personnel

Nothing to report.

Section VI. Project Output

Patents:

Nothing to report.

Publications / Presentations:

Cole, G. M., Nazari, A., Green, M., Vermaas, W., Quinn, J. C., Economic and life cycle assessment of direct air capture for algal systems using moisture swing sorbents, Algal Biomass Biofuels and Bioproducts [Conference Session), Waikoloa Beach, HI, June 13, 2023.

Other

Invention disclosure submitted to ASU Skysong Innovations on 6/23/2023: Vermaas, Wim, Flory, Justin, Li, Shuqin, Green, Matthew, Velazco Medel, Marlene Alejandra, Lackner, Klaus, Taylor, Samantha, Methods and systems for moisture-driven carbon dioxide capture and delivery to photosynthetic organisms, Tech ID: M24-006P^